# Case study - Two tier Highly Available and Highly Scalable architecture.

# Problem statement

**Service Covered:** VPC, EC2, RDS, S3, ELB, CloudWatch, ASG, SNS, CloudFront, AWS WAF, System Manager

**Learning objective:**

- Articulate the characteristics of the AWS Cloud Computing Platform.
- Describe the core benefits of using AWS services.
- Compare and contrast AWS cloud services to On-Premises Infrastructure.
- Implement hosting a multi-tier highly available and scalable architecture using AWS services.

**Expectations from the Participants**

- Participants will prepare a word document listing the benefits of Cloud Computing.
- Participants will then do a team discussion to compare On-Premises and cloud services.
- Participants will create a multi-tier highly available and scalable architecture using AWS services.
- Participants will prepare a word document listing the screenshot of case study implementation.

**Background:**

Cloudforall.in is an India based manufacturing firm with a global presence, collaborating with partner entities to launch products. Cloudforall.in produces goods based on proprietary automobile company requirement.

**Existing Environment**

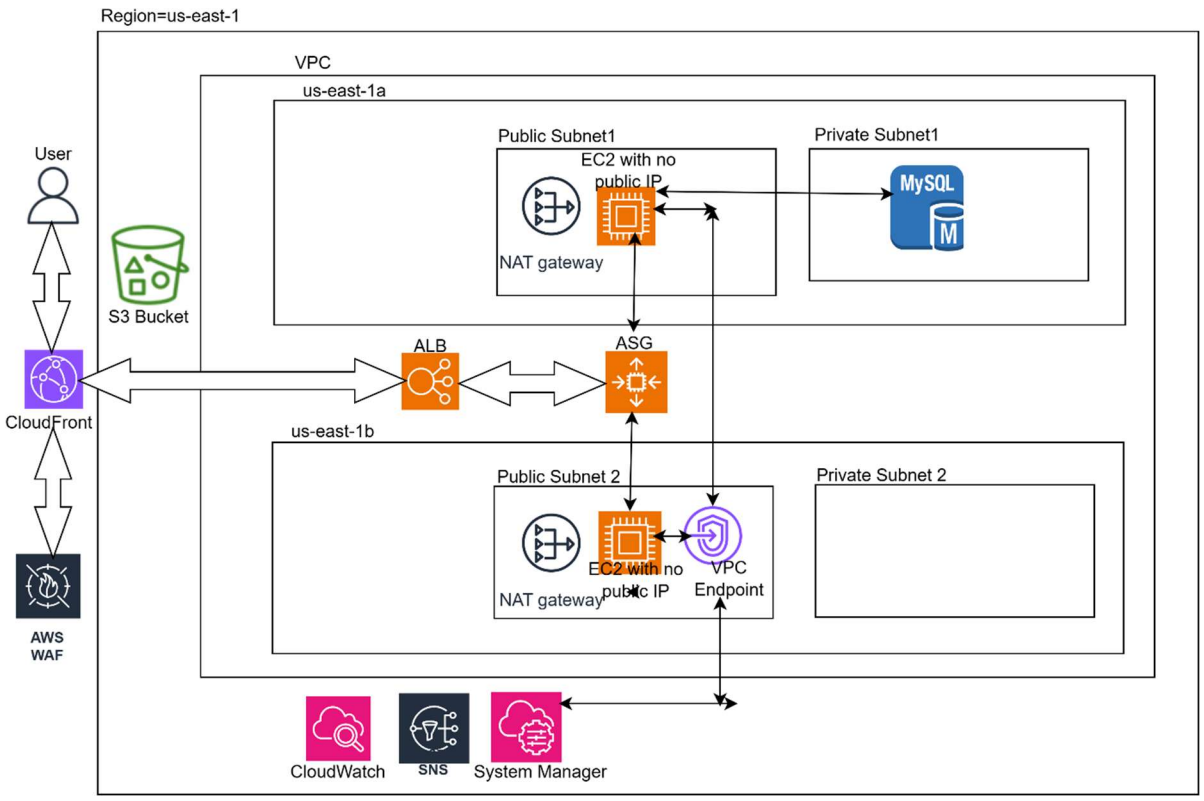
For its business marketing, Cloudforall currently uses a variety of server type hosted in their data center, including web servers, and Database Server. There are on-premises load balancers that are also connected to webservers to give high availability for the application. This entire setup is managed by company employees which need huge capex and opex to the company.

**Requirements –**

Cloudforall intends to make the following infrastructure modifications:

- Transfer webapp whole tier set to AWS without public IP.
- Transfer database whole tier set to AWS in a secure network environment.
- Make sure webapp will be high available and scalable.
- Secure the webapp from DDoS attack.
- Provide low latency to the webapp.
- Make sure webapp servers get updated on regular basis.
- Monitor the webapp server CPU utilization closely.
- Make sure images on webpages get served from cost-effective storage.
- Make sure operation team will get notified if any webapp server is failed to launch.

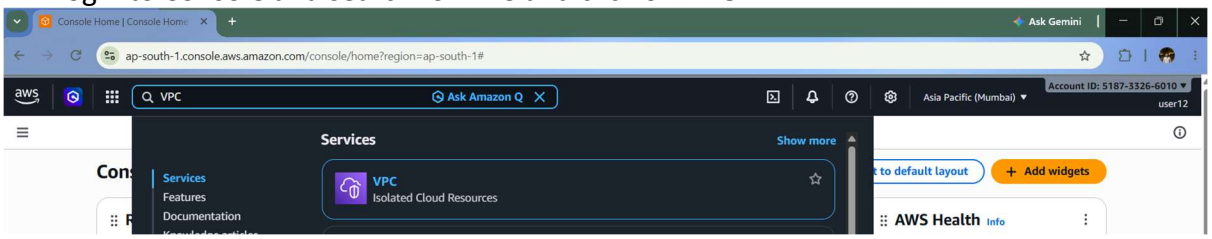
# Architecture Diagram:



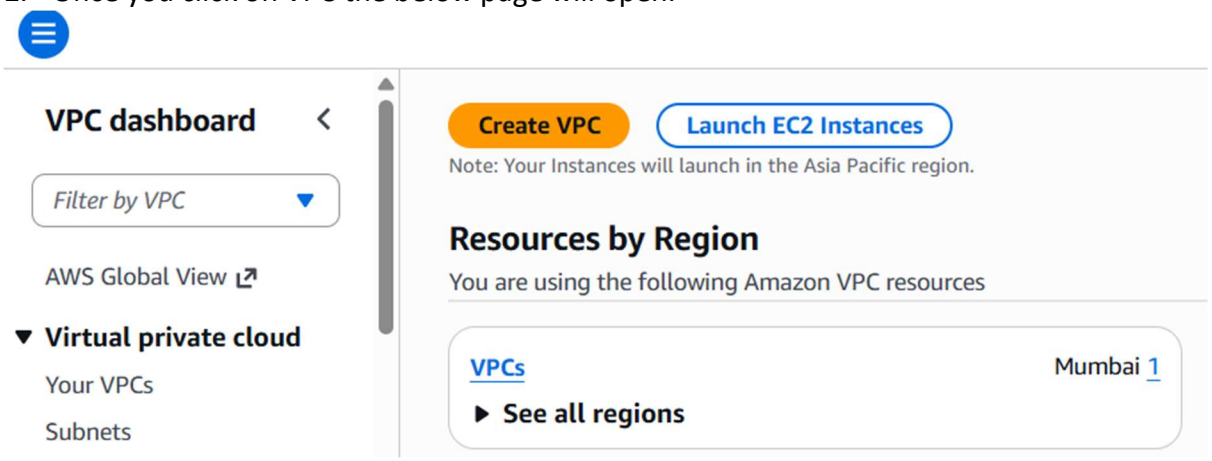
# Solution

## Task 1: Create a VPC

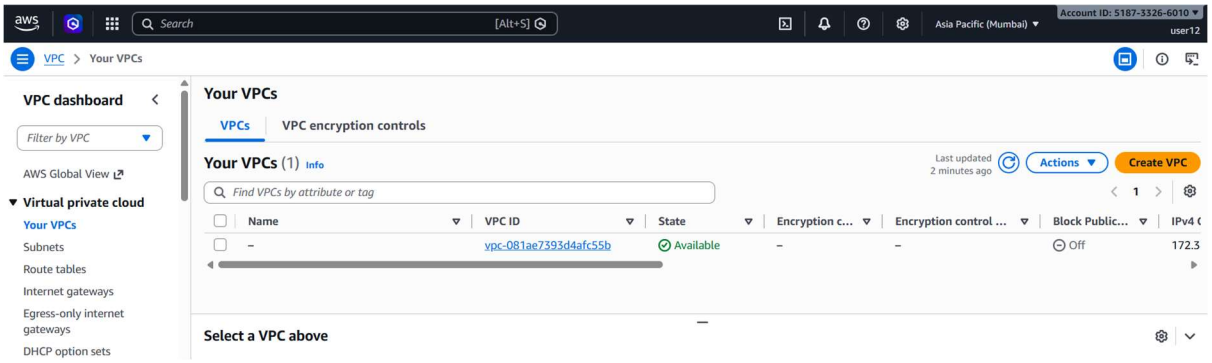
1. Login to Console and search for VPC and click on **VPC**.



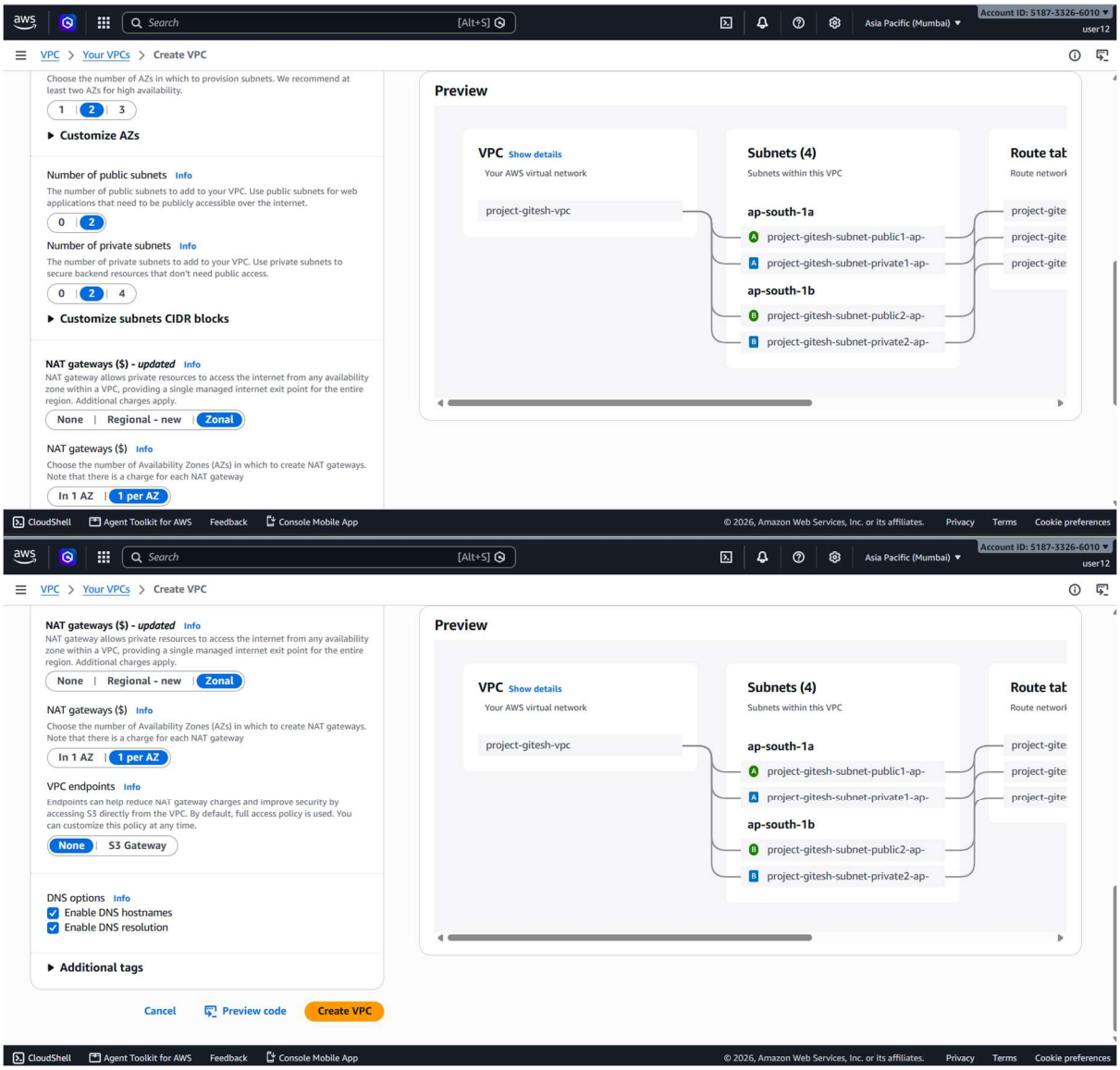

2. Once you click on VPC the below page will open.

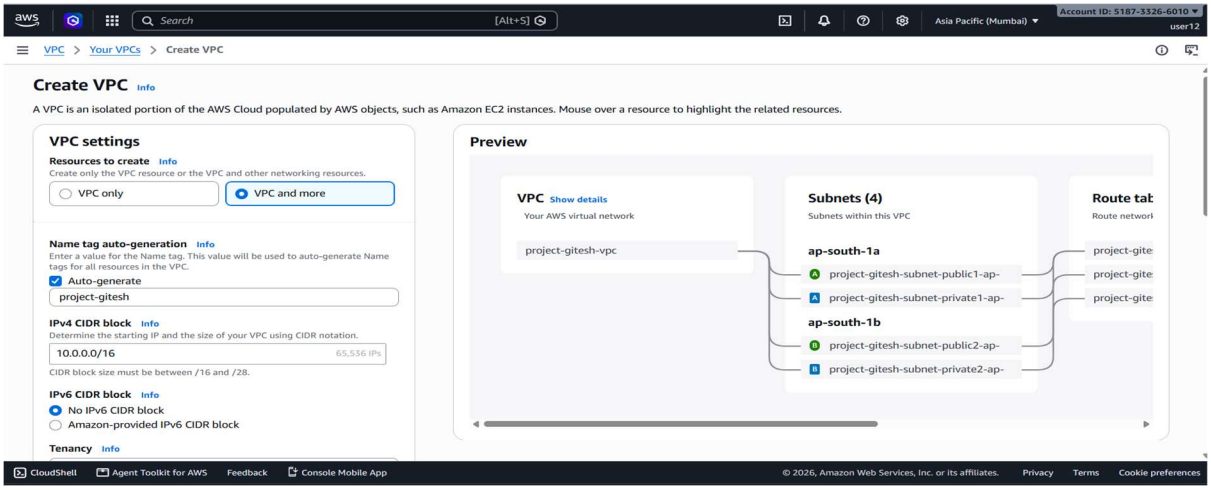



3. Click on **VPC** and then click on **Create VPC**.



4. Once you click on **Create VPC**, the below page will appear. Click as per the below highlights that will create a **VPC**, **CIDR blocks**, **2 Private Subnets**, **2 Public Subnets**, **NAT Gateway** per public subnet, **Internet Gateway**, **Route Table**, Entries in the Route table will get automatically updated.
5. Then click on the **Create VPC**.

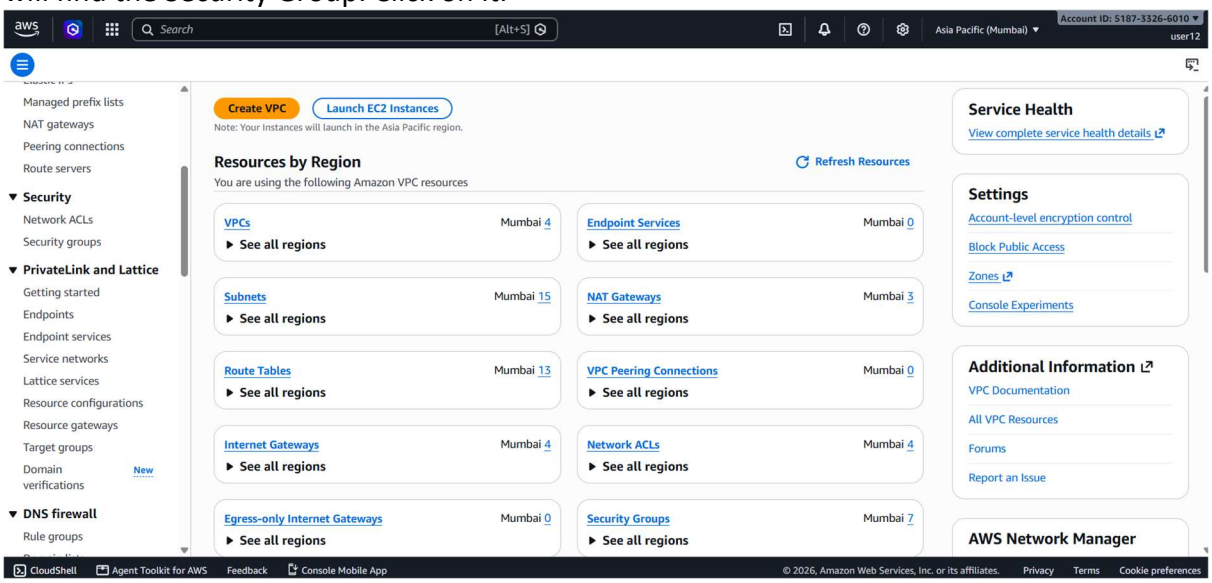




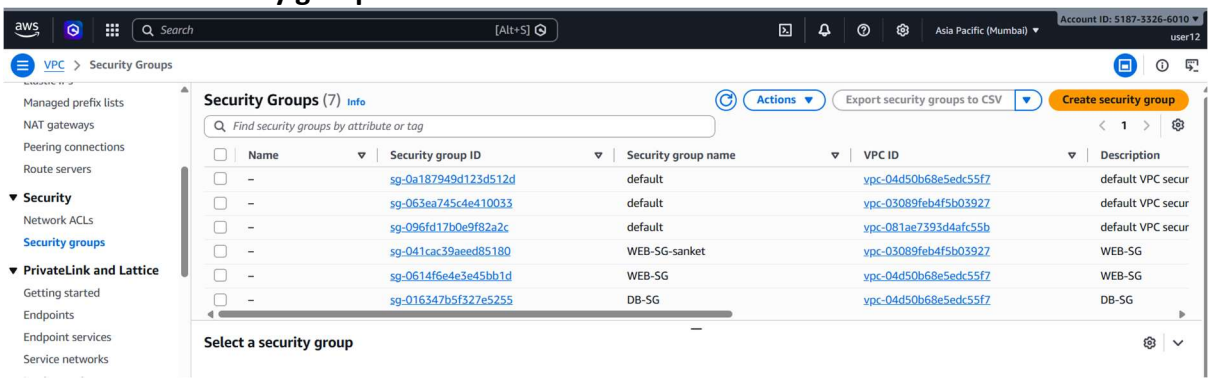

This will create your VPC.

# Task 2: Create 2 Security Groups

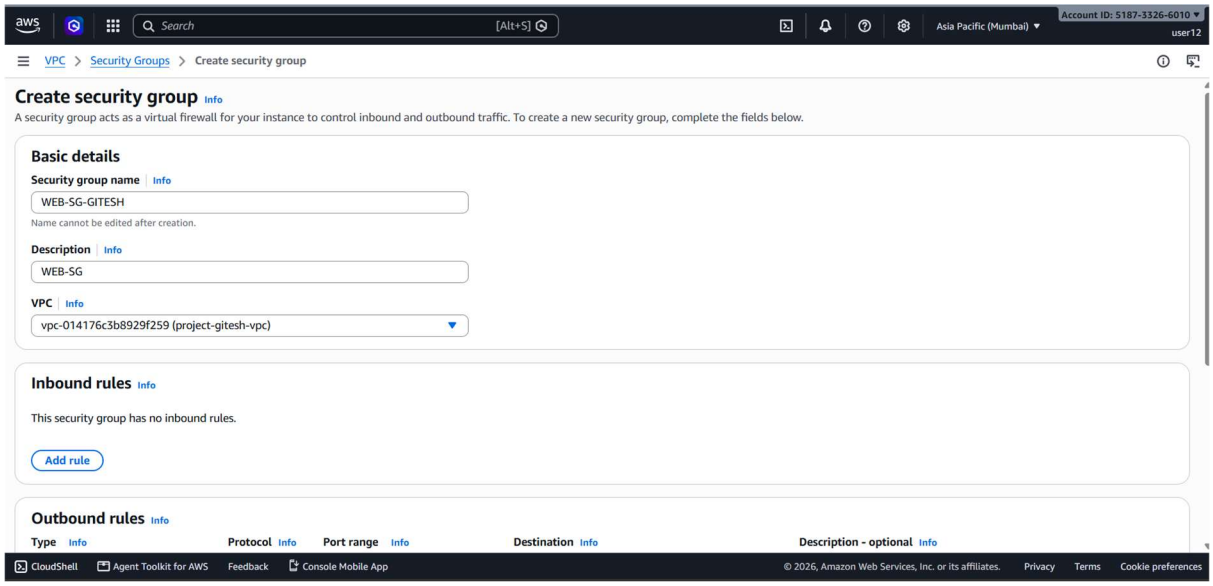
1. Go to the AWS Management console and search for VPC and in the left-hand side we will find the Security Group. Click on it.



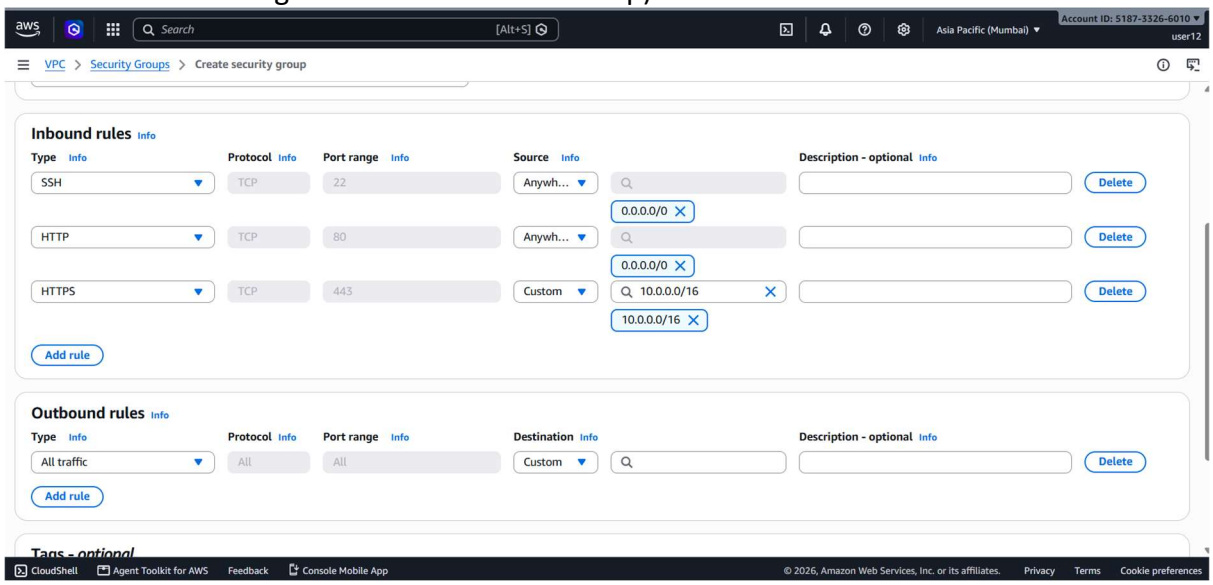

2. Select **Create security group.**



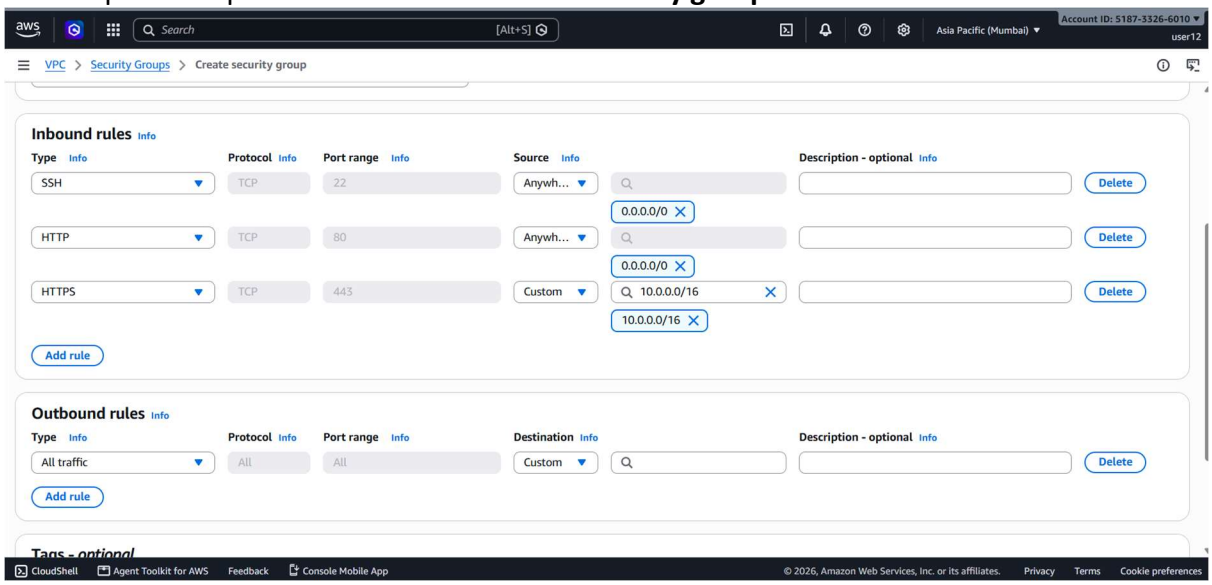
3. Give name and description as WEB-SG and select demo-vpc created in step 1. In the inbound rule section click on **Add rule.**




4. Add rule as shown below (for testing instance we are opening ssh and http for 0.0.0.0/0, later on we will change source IP for ssh and http)

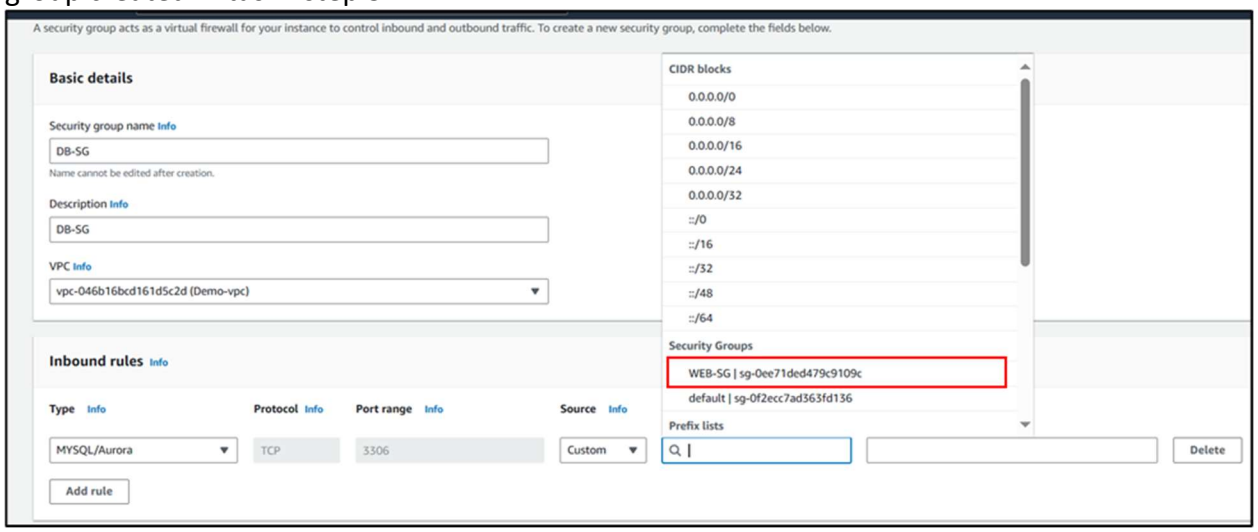


5. Other option keep default and select **create security group.**

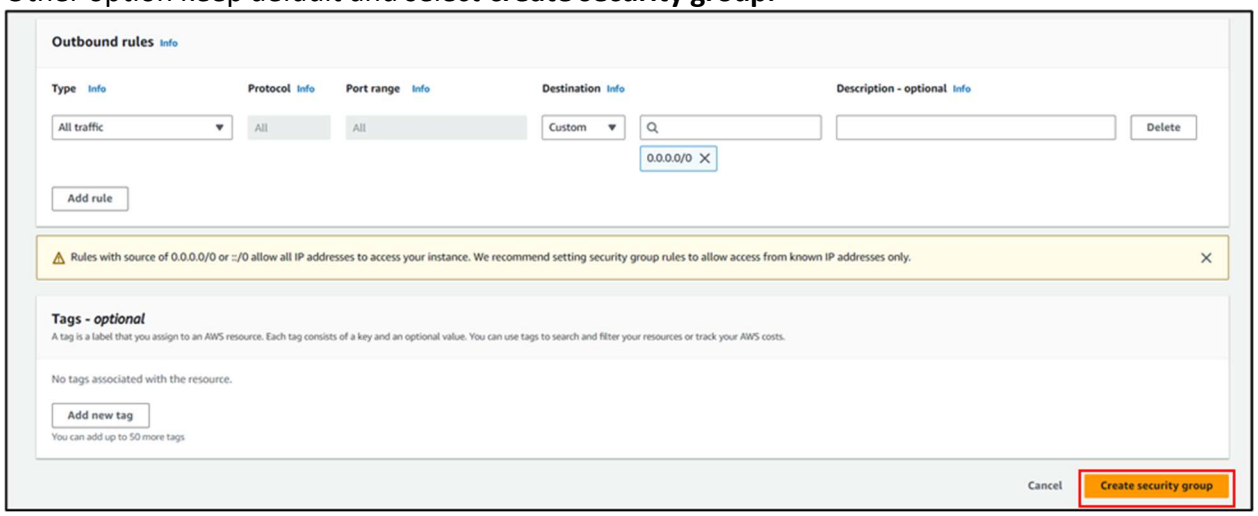



6. Create one more security group for database. Go to VPC service and select security group from left hand side then click on create security group. Give name and description as DB-SG and select demo-vpc created in step 1. In the inbound rule section click on **Add rule.**

7. In Type select **MYSQL/Aurora**, in source select **Custom** and click on **WEB-SG** (security group created in task2 step 3.



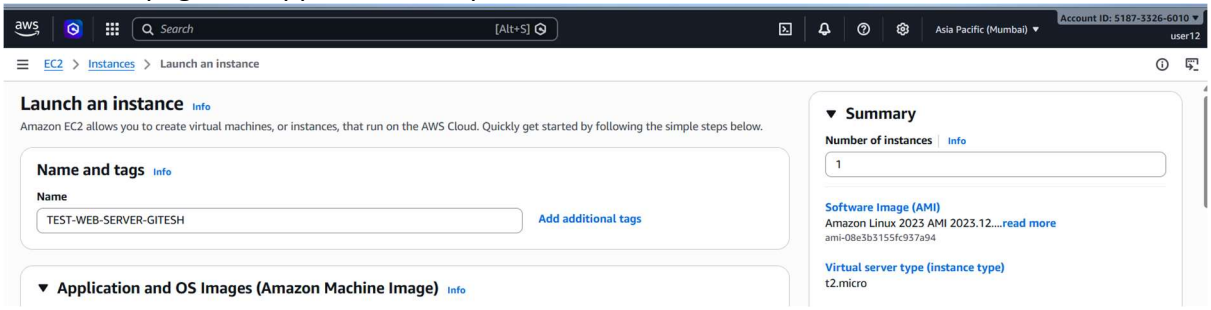
8. Other option keep default and select **create security group.**



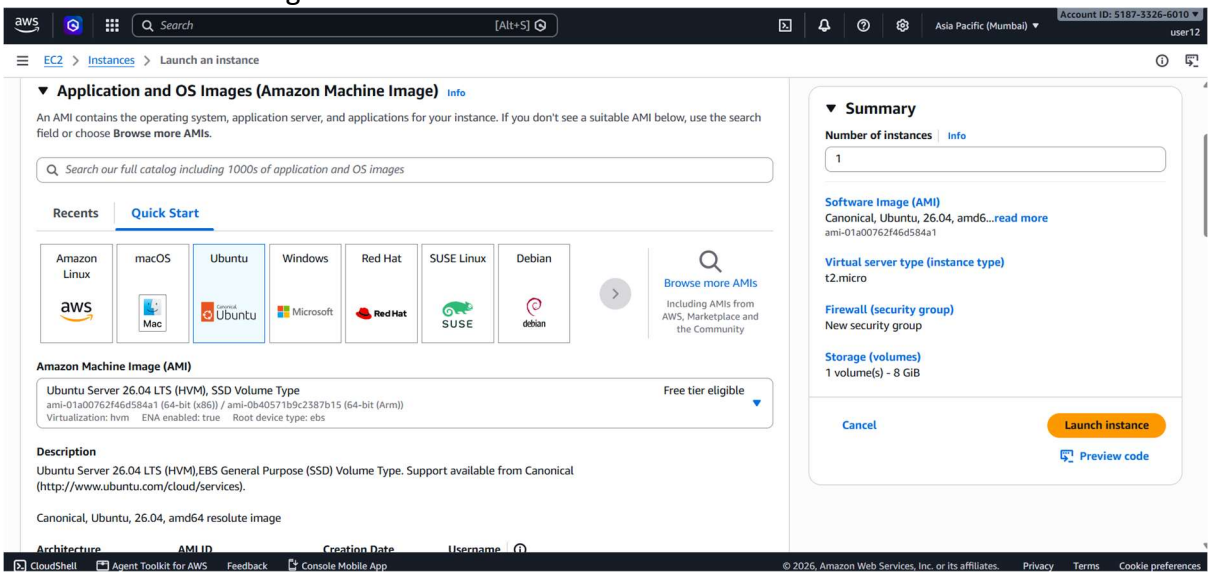
# Task 3: Create, connect, and configure Instance

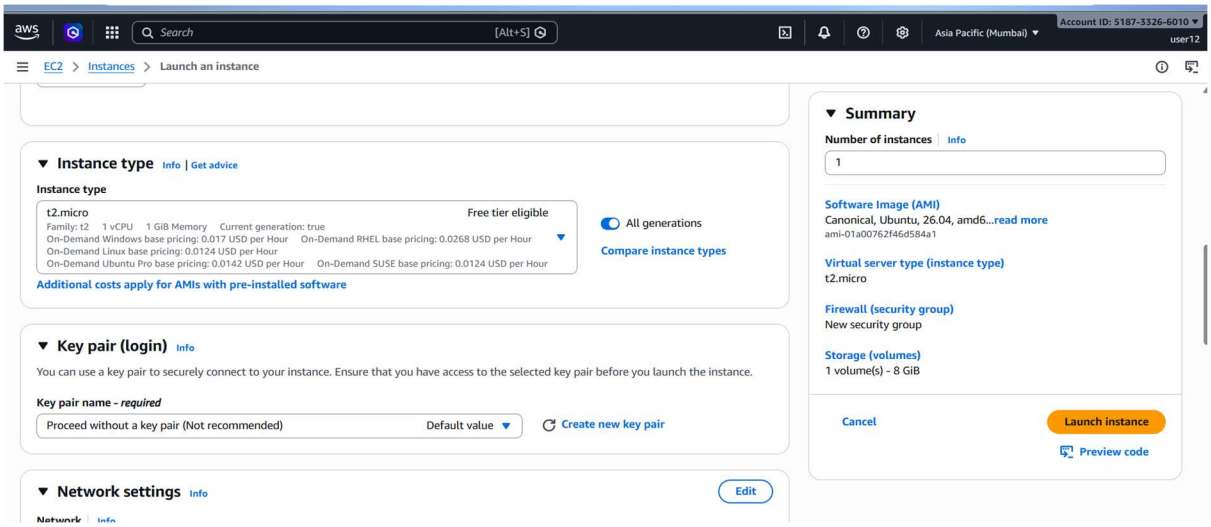
Now we are supposed to launch 1 testing Instance which we will use to install our require packages to host the application, also we will test whether application running on this instance is connecting to RDS databases successfully and finally we will create our AMI image from it.

1. Go to console and search for EC2 and click on EC2.
2. Then click on instances (running).
3. Click on launch instances.
4. The below page will appear. Please put the instance name TEST-WEB-SERVER.

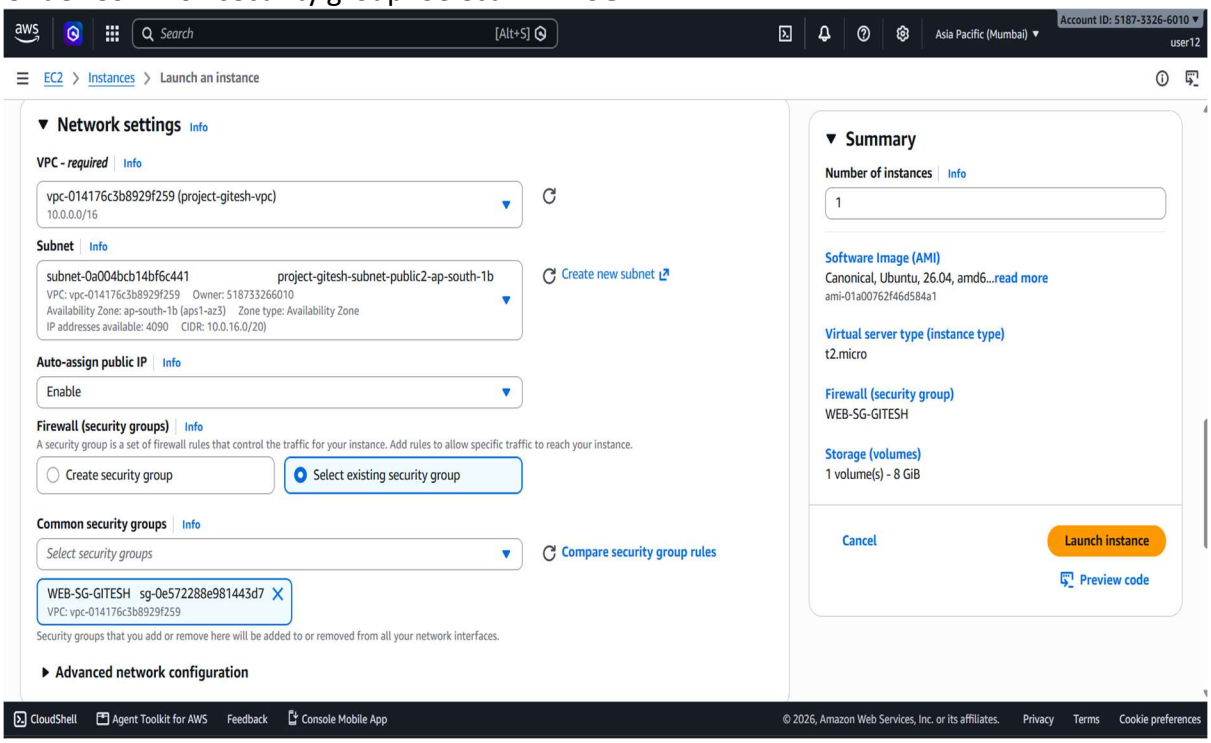



5. Select the Ubuntu AMI
   a) Under instance type – change it to Amazon Linux 2 AMI (HVM)
   b) Under Key pair (login) – Select Proceed without a key pair (not recommended)
   c) Under Network settings – select edit.

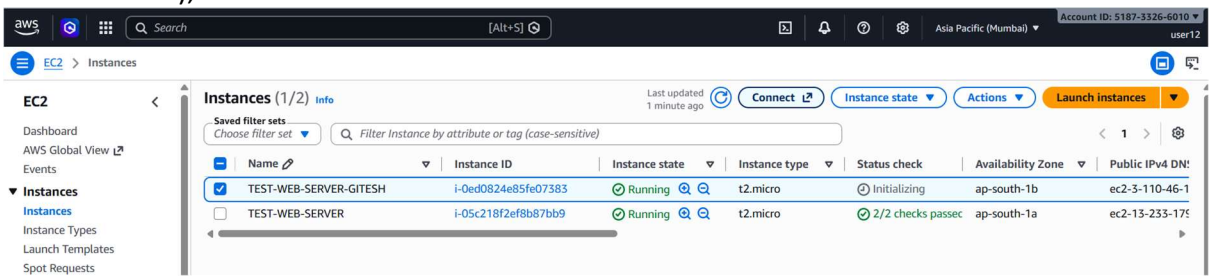




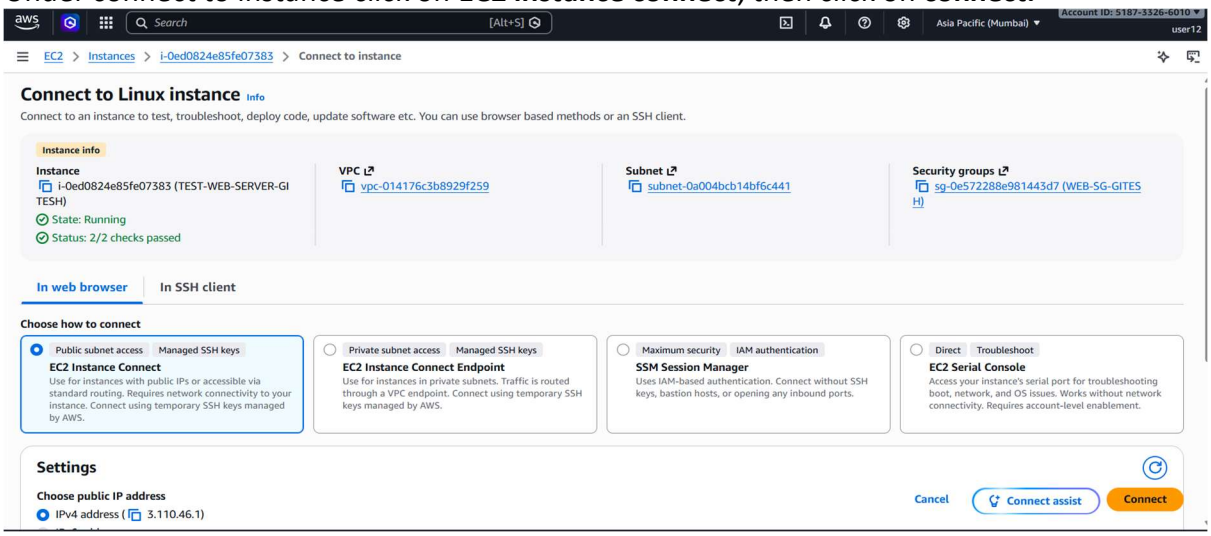

6. When you click on edit the below page will get displayed. We are supposed to select the VPC which we have created with name **Demo-vpc.**
   a) Under subnets we must select any one **public subnet** that we have created previously.
   b) Under Auto assign public IP select enable.
   c) Under Firewall (Security Groups) – Select **existing security group.**
   d) Under Common security group- Select **WEB-SG**.



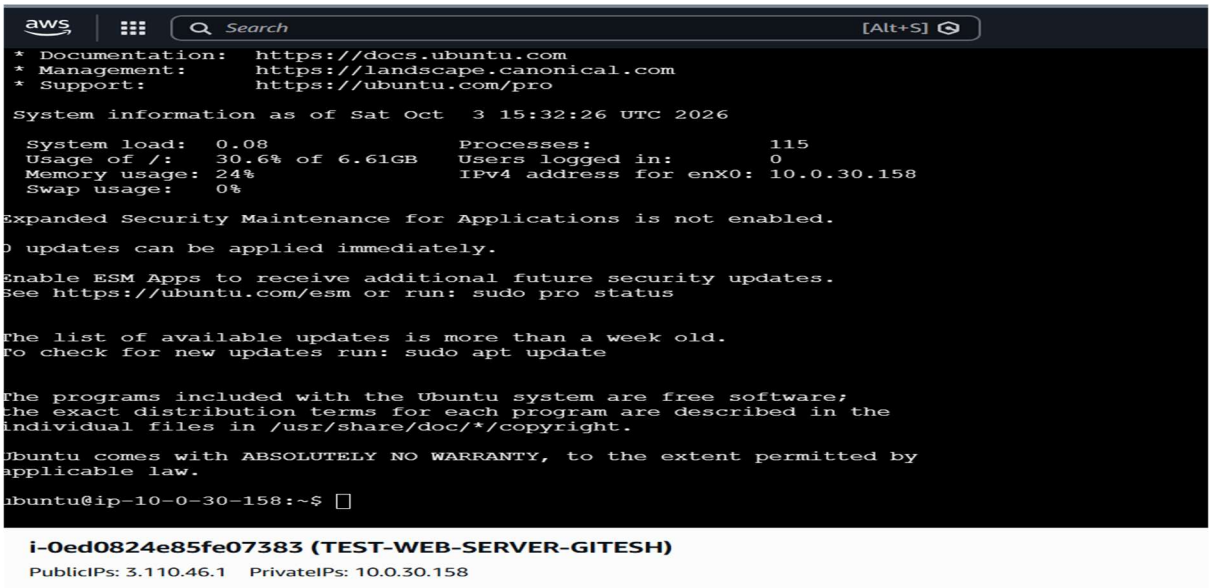
7. Rest keeps it as it is and select **Launch instance**.
8. Once instance launch successfully, connect to the instance from console. For this go to EC2 service, select **instances** option from left hand side, click on your instance (**TEST-WEB-SERVER**), select **connect.**




9. Under connect to instance click on **EC2 instance connect**, then click on **connect.**

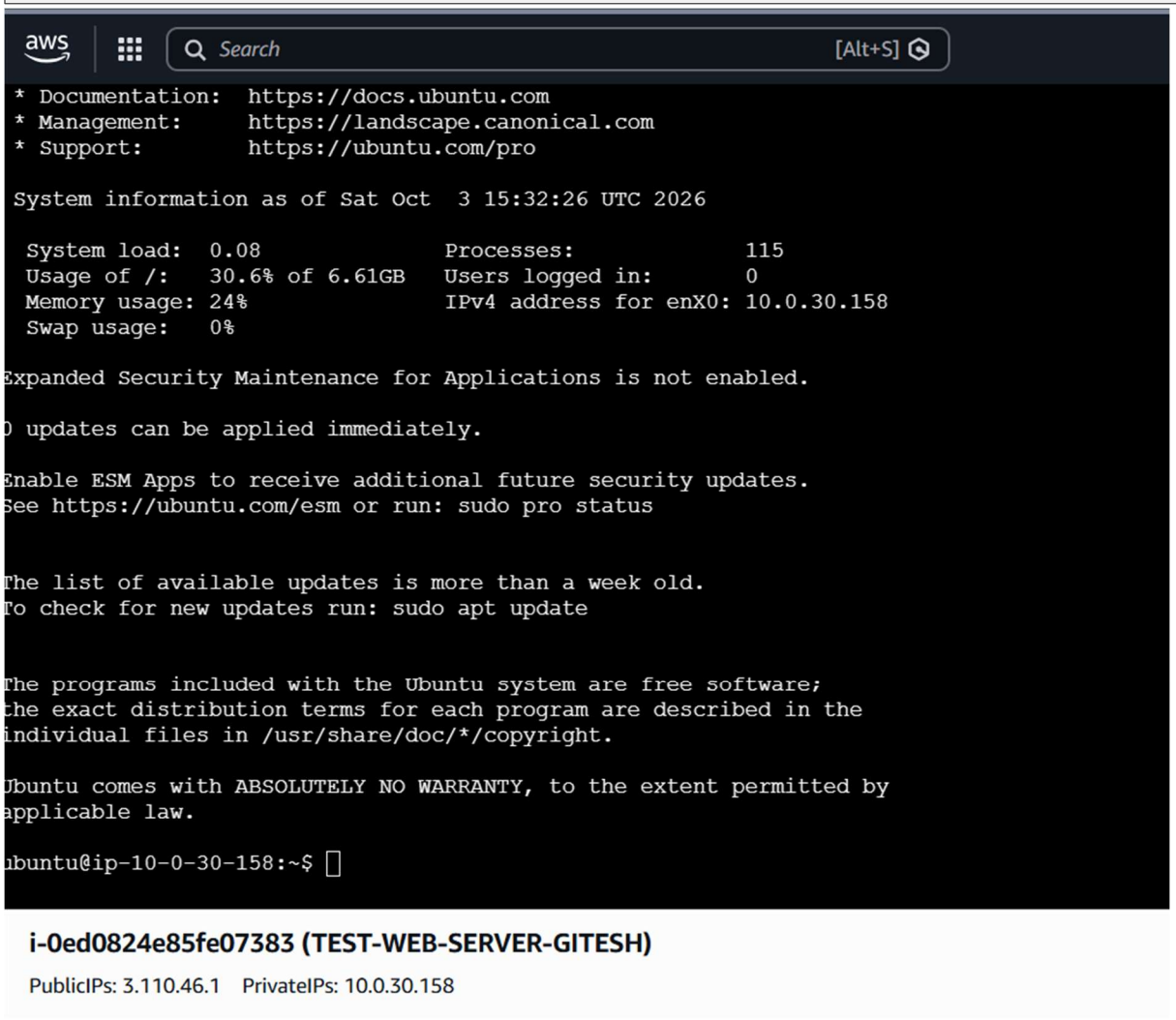



10. A new tab will get open your browser which will give you cl access to your ubuntu instance.



11. Put the below commands so that the instances will get updated first.

```
$ sudo su
$ apt-get update -y ( If this command gives you error then you can use apt update command)
```

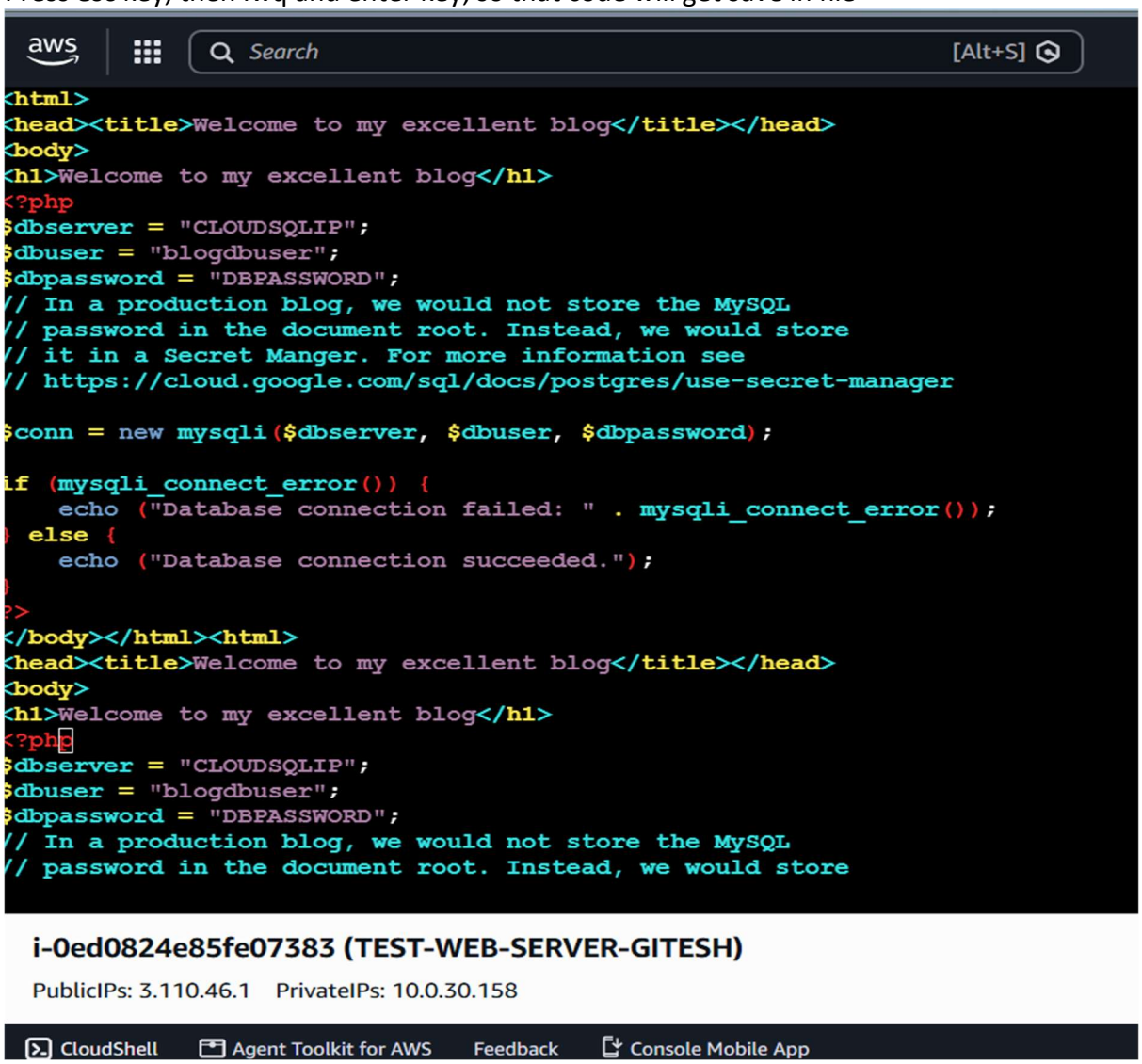


12. Now give the below command one by one

$ apt-get install apache2 php php-mysql -y
$ service apache2 restart
$ service apache2 enable
$ service apache2 status
$ cd /var/www/html $ vi index.php

13. New blank window will open for you. Copy the below code without any modifications in blank window. (Before pasting the code press, I key to insert data in blank file).

14. Press **esc** key, then :wq and enter key, so that code will get save in file

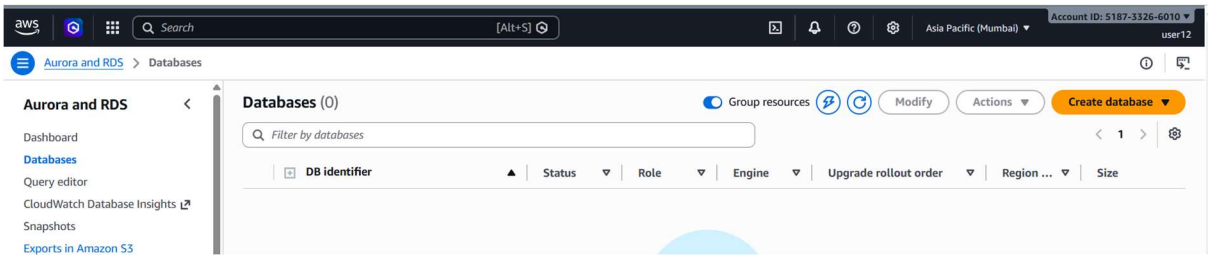


15. Now copy and paste the public IP of your instance in browser and add /index.php after the public IP of your instance. Below is the expected output. (Note that it is not displaying the message that database is connected successfully). Now next task is to create database and connect to your web application.

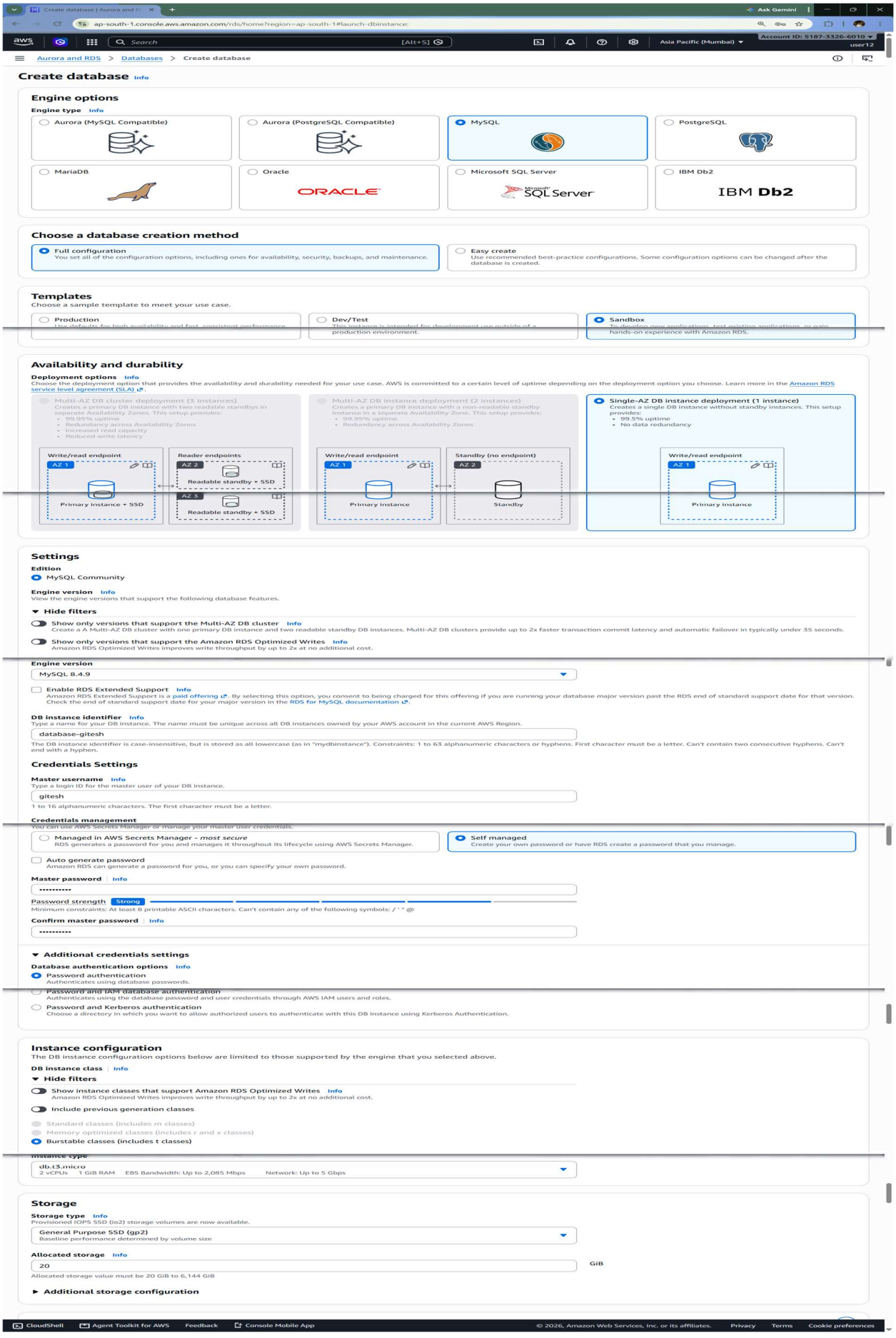


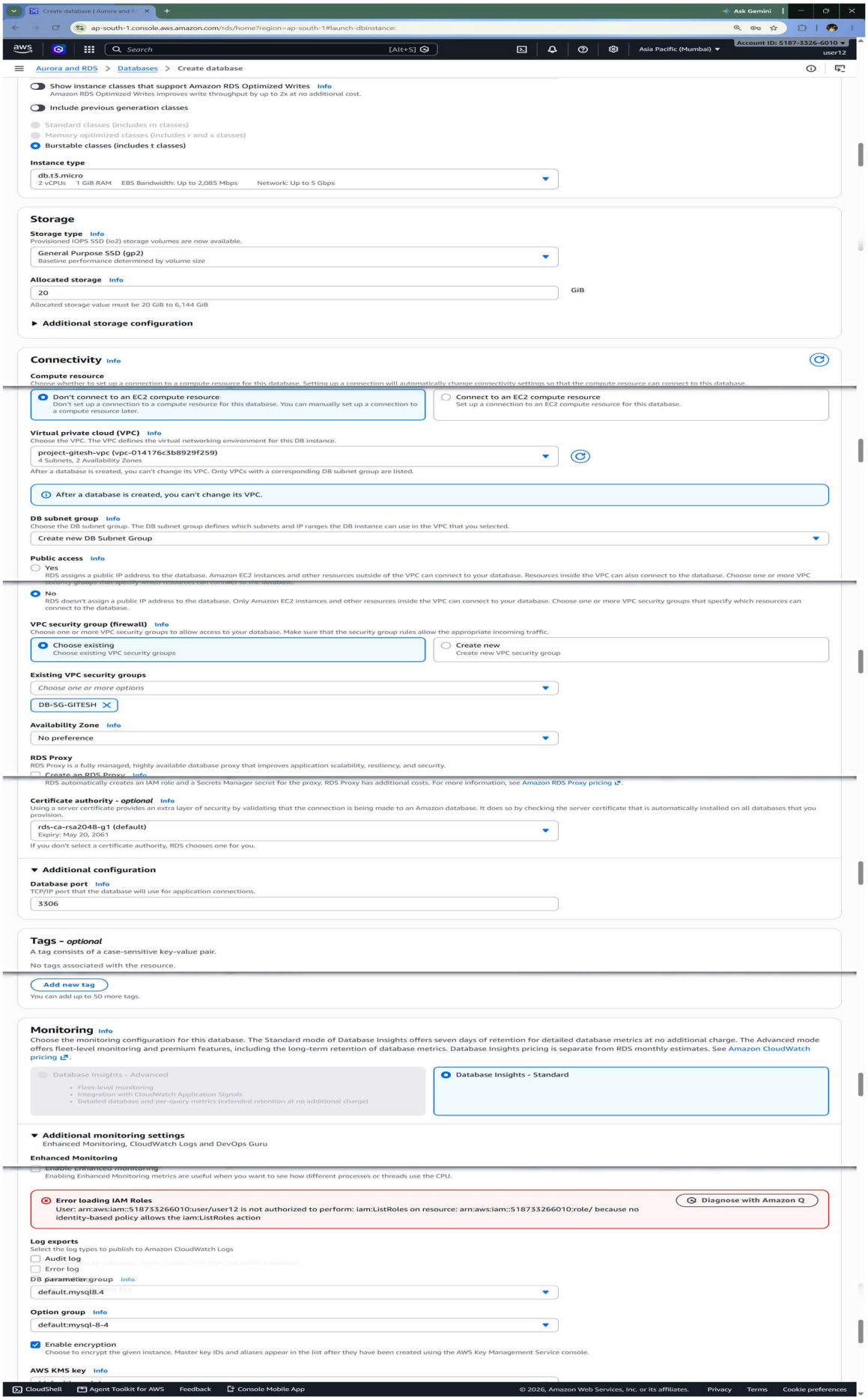
# Task 4: Create, RDS MySQL database.

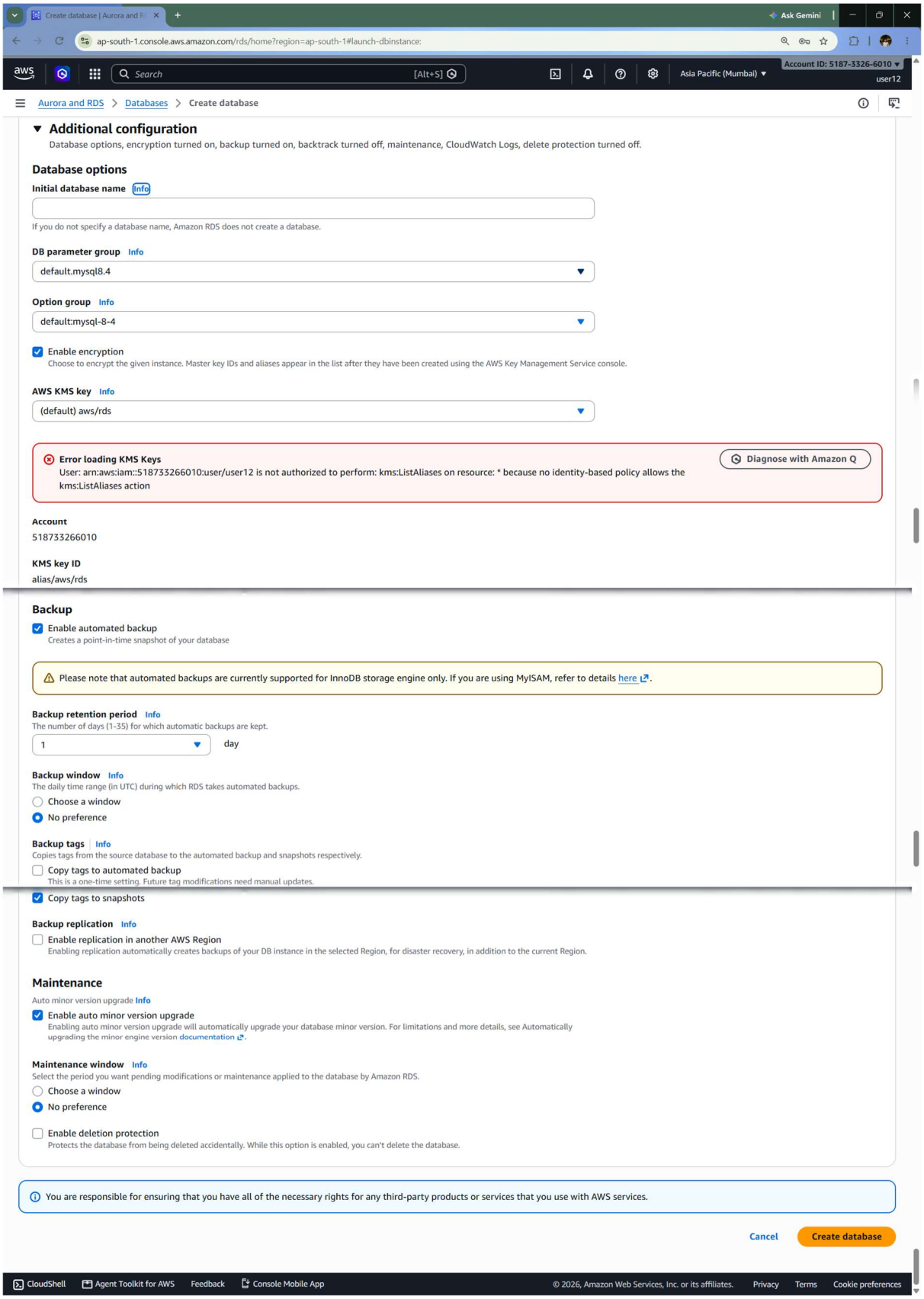
1. Select **Database** then click on **Create Database**




2. Select the **Standard Create,** Under Engine options select **MySQL.**
3. Under Template select **Free Tier**
4. Under setting Provide the DB identifier then provide the **username** and **password** for Database
5. Keep Instance configuration and storage setting default.
6. Under **Connectivity** Select **Demo VPC** and select subnet group
7. Select **No** for public access setting.
8. In **VPC security group** setting choose existing security group and select **DB-SG** Security group
9. Expand Advance setting and disable **automated backup** option.
10. Leave other setting as it is then click on **Create database.**


Create database | Aurora and R
Ask Gemini
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Alt+S]
Asia Pacific (Mumbai)
Account ID: 5187-3326-6010
user12
Aurora and RDS > Databases > Create database
Create database Info
Engine options
Engine type Info
Aurora (MySQL Compatible)
Aurora (PostgreSQL Compatible)
MySQL
PostgreSQL
MariaDB
Oracle
Microsoft SQL Server
IBM Db2
Choose a database creation method
Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Templates
Choose a sample template to meet your use case.
Production
Use defaults for high availability and fast, consistent performance.
Dev/Test
This instance is intended for development use outside of a production environment.
Sandbox
To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.
Availability and durability
Deployment options Info
Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the Amazon RDS service level agreement (SLA).
Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
99.95% uptime
Redundancy across Availability Zones
Increased read capacity
Reduced write latency
Write/read endpoint
AZ 1
Primary instance + SSD
Reader endpoints
AZ 2
Readable standby + SSD
AZ 3
Readable standby + SSD
Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
99.95% uptime
Redundancy across Availability Zones
Write/read endpoint
AZ 1
Primary instance
Standby (no endpoint)
AZ 2
Standby
Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:
99.5% uptime
No data redundancy
Write/read endpoint
AZ 1
Primary instance
Settings
Edition
MySQL Community
Engine version Info
View the engine versions that support the following database features.
Hide filters
Show only versions that support the Multi-AZ DB cluster Info
Create a A Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.
Show only versions that support the Amazon RDS Optimized Writes Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
Engine version
MySQL 8.4.9
Enable RDS Extended Support Info
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the RDS for MySQL documentation.
DB instance identifier Info
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-gitesh
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.
Credentials Settings
Master username Info
Type a login ID for the master user of your DB instance.
gitesh
1 to 16 alphanumeric characters. The first character must be a letter.
Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
Self managed
Create your own password or have RDS create a password that you manage.
Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.
Master password Info
Password strength Strong
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @
Confirm master password Info
Additional credentials settings
Database authentication options Info
Password authentication
Authenticates using database passwords.
Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.
Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.
DB instance class Info
Hide filters
Show instance classes that support Amazon RDS Optimized Writes Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
Include previous generation classes
Standard classes (includes m classes)
Memory optimized classes (includes r and x classes)
Burstable classes (includes t classes)
Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps
Storage
Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size
Allocated storage Info
20
GiB
Allocated storage value must be 20 GiB to 6,144 GiB
Additional storage configuration
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

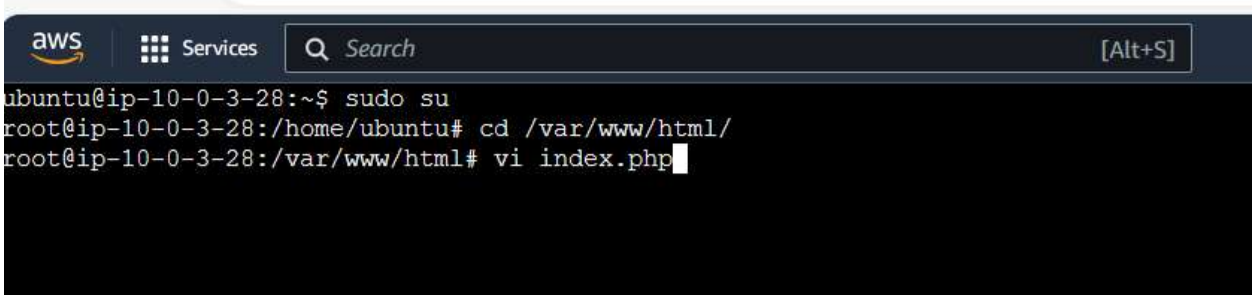

Create database | Aurora and R
Ask Gemini
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Alt+S]
Asia Pacific (Mumbai)
Account ID: 5187-3326-6010
user12
Aurora and RDS > Databases > Create database
Show instance classes that support Amazon RDS Optimized Writes Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
Include previous generation classes
Standard classes (includes m classes)
Memory optimized classes (includes r and x classes)
Burstable classes (includes t classes)
Instance type
db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps
Storage
Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size
Allocated storage Info
20
GiB
Allocated storage value must be 20 GiB to 6,144 GiB
Additional storage configuration
Connectivity Info
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
project-gitesh-vpc (vpc-014176c3b8929f259)
4 Subnets, 2 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
Create new DB Subnet Group
Public access Info
Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Existing VPC security groups
Choose one or more options
DB-SG-GITESH
Availability Zone Info
No preference
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.
Certificate authority - optional Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061
If you don't select a certificate authority, RDS chooses one for you.
Additional configuration
Database port Info
TCP/IP port that the database will use for application connections.
3306
Tags - optional
A tag consists of a case-sensitive key-value pair.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Monitoring Info
Choose the monitoring configuration for this database. The Standard mode of Database Insights offers seven days of retention for detailed database metrics at no additional charge. The Advanced mode offers fleet-level monitoring and premium features, including the long-term retention of database metrics. Database Insights pricing is separate from RDS monthly estimates. See Amazon CloudWatch pricing.
Database Insights - Advanced
Fleet-level monitoring
Integration with CloudWatch Application Signals
Detailed database and per-query metrics (extended retention at no additional charge)
Database Insights - Standard
Additional monitoring settings
Enhanced Monitoring, CloudWatch Logs and DevOps Guru
Enhanced Monitoring
Enable Enhanced monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.
Error loading IAM Roles
User: arn:aws:iam::518733266010:user/user12 is not authorized to perform: iam:ListRoles on resource: arn:aws:iam::518733266010:role/ because no identity-based policy allows the iam:ListRoles action
Diagnose with Amazon Q
Log exports
Select the log types to publish to Amazon CloudWatch Logs
Audit log
Error log
DB parameter group Info
default.mysql8.4
Option group Info
default:mysql-8-4
Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console.
AWS KMS key Info
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Create database | Aurora and R
Ask Gemini
ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:
Search
[Alt+S]
Asia Pacific (Mumbai)
Account ID: 5187-3326-6010
user12
Aurora and RDS
Databases
Create database
Additional configuration
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.
Database options
Initial database name Info
If you do not specify a database name, Amazon RDS does not create a database.
DB parameter group Info
default.mysql8.4
Option group Info
default:mysql-8-4
Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console.
AWS KMS key Info
(default) aws/rds
Error loading KMS Keys
User: arn:aws:iam::518733266010:user/user12 is not authorized to perform: kms:ListAliases on resource: * because no identity-based policy allows the kms:ListAliases action
Diagnose with Amazon Q
Account
518733266010
KMS key ID
alias/aws/rds
Backup
Enable automated backup
Creates a point-in-time snapshot of your database
Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MyISAM, refer to details here.
Backup retention period Info
The number of days (1-35) for which automatic backups are kept.
1
day
Backup window Info
The daily time range (in UTC) during which RDS takes automated backups.
Choose a window
No preference
Backup tags Info
Copies tags from the source database to the automated backup and snapshots respectively.
Copy tags to automated backup
This is a one-time setting. Future tag modifications need manual updates.
Copy tags to snapshots
Backup replication Info
Enable replication in another AWS Region
Enabling replication automatically creates backups of your DB instance in the selected Region, for disaster recovery, in addition to the current Region.
Maintenance
Auto minor version upgrade Info
Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation.
Maintenance window Info
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.
Choose a window
No preference
Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.
You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.
Cancel
Create database
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

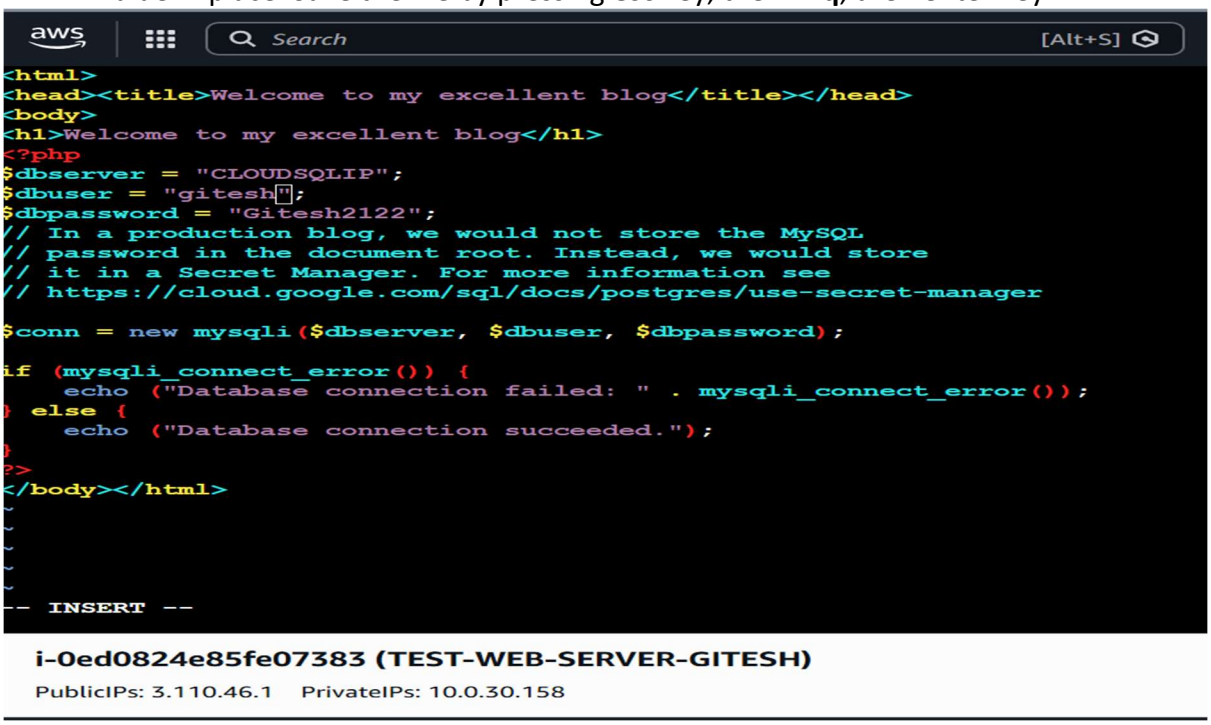
# Task 5: Configure Instance to connect to MySQL Database

1. Connect to the instance created in task one. Give below command.

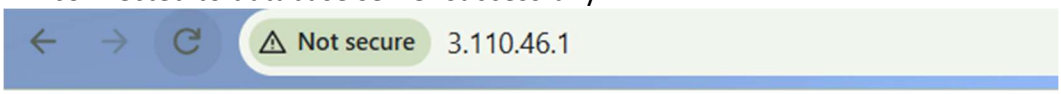
```
$ sudo su
$ cd /var/www/html
$ vi index.php
```




2. In the vi text editor, press "**insert**" key, replace **CLOUDSQLIP** with the database endpoint that you created in task 3. Leave the quotation marks around the value in place. Replace **DBPASSWORD** with the MySQL database password that you defined in task 3. Leave the quotation marks around the value in place. Replace **blogdbuser** with the MySQL database password that you defined in task 3. Leave the quotation marks around the value in place. Save the file by pressing **esc** key, then :**wq**, then enter key.



3. Now copy and paste the public IP of your instance in browser and add /index.php after the public IP of your instance. Below is the expected output. (Note that it is now showing that database is connected successfully). It means that application server is connected to database server successfully.

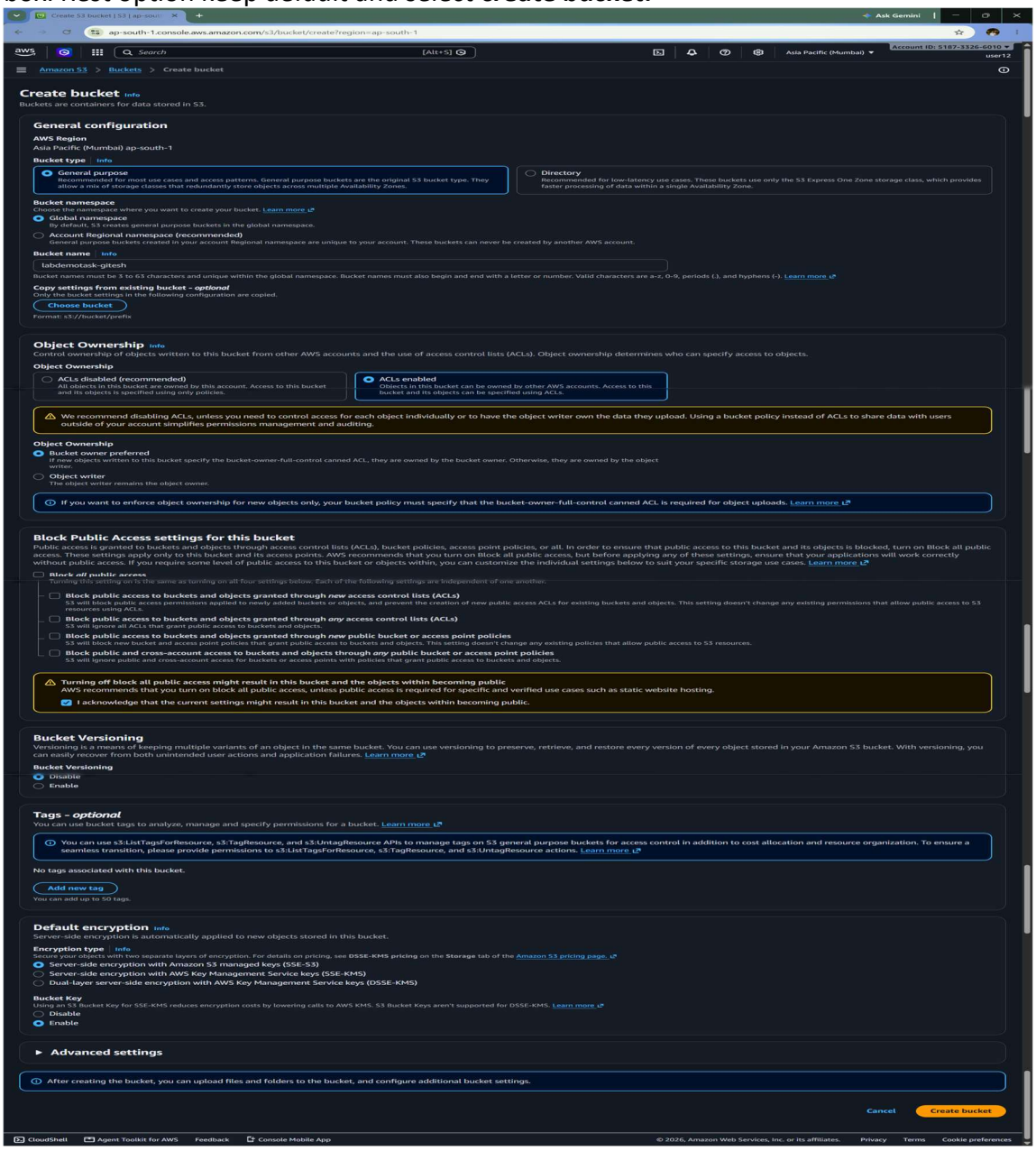



# Welcome to my excellent blog

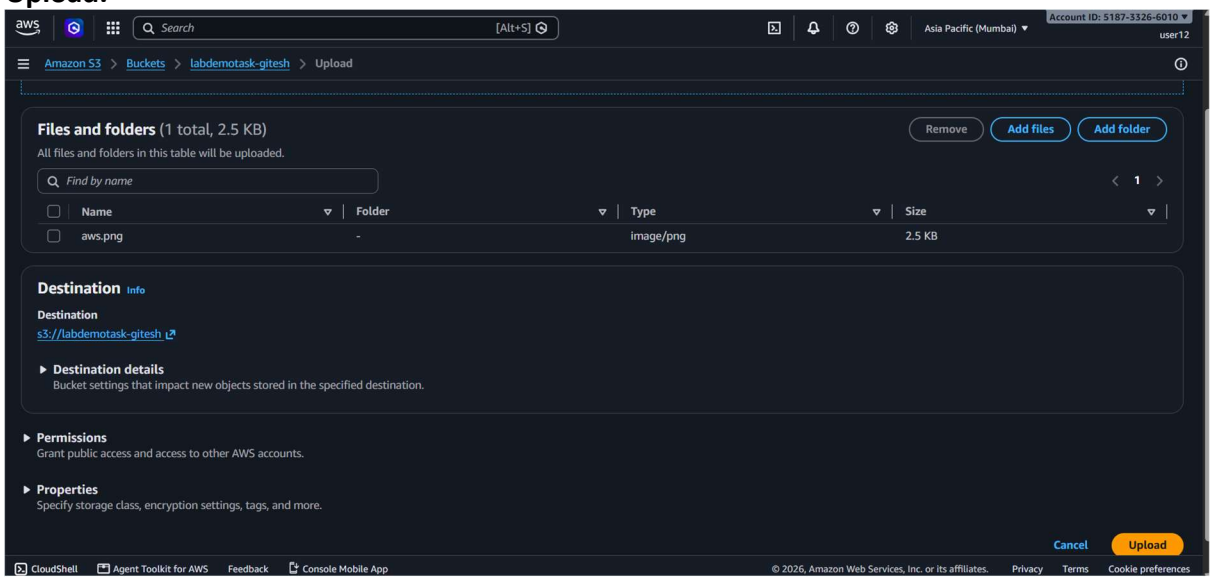
Database connection succeeded.

# Task 6: Configure Instance to get the images from Amazon S3 on website.

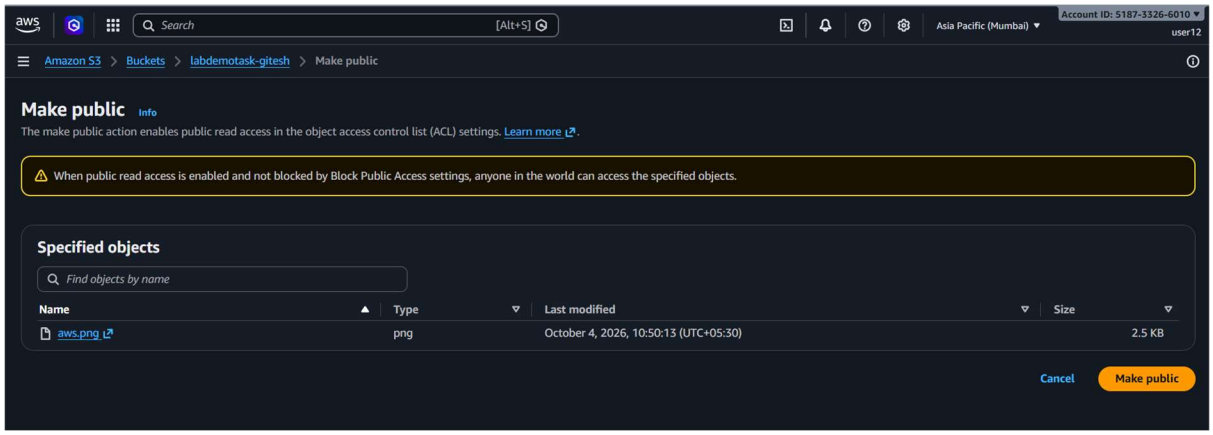
1. Go to AWS Management Console. In Search, Type S3
2. Click on Create Bucket, Give the Bucket name as "**labdemotask-1**" (Bucket name should be globally Unique)
3. Select the region as instructed by instructor, in **ACL ownership** section select **ACL enabled.**
4. In **Block public access setting,** disable block public access and enable **I acknowledge** check box. Rest option keep default and select **create bucket.**



5. Open newly created bucket. Select **object**, click on **upload**. Click on **add file.** Select one picture image from your laptop and then click **Ok.** Keep other option default and click **Upload.**

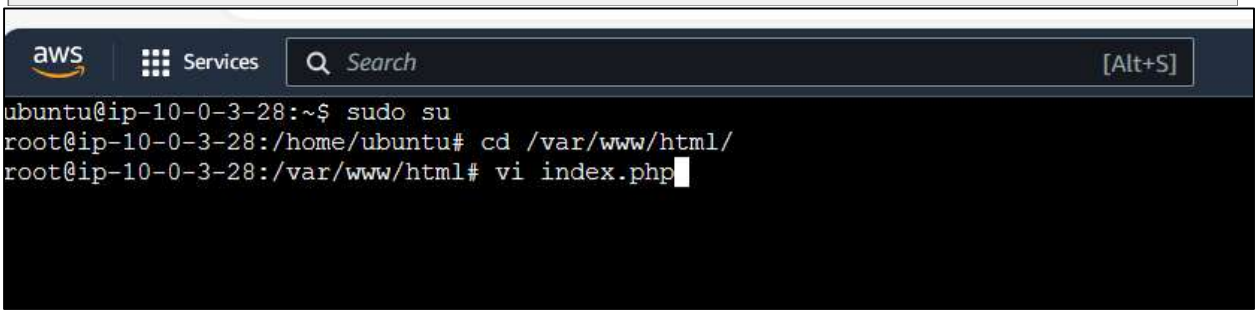



6. Select the object, which was uploaded in last step, go to **Action** and click on **Make public using ACL.**
7. Select Make Public and click on close.

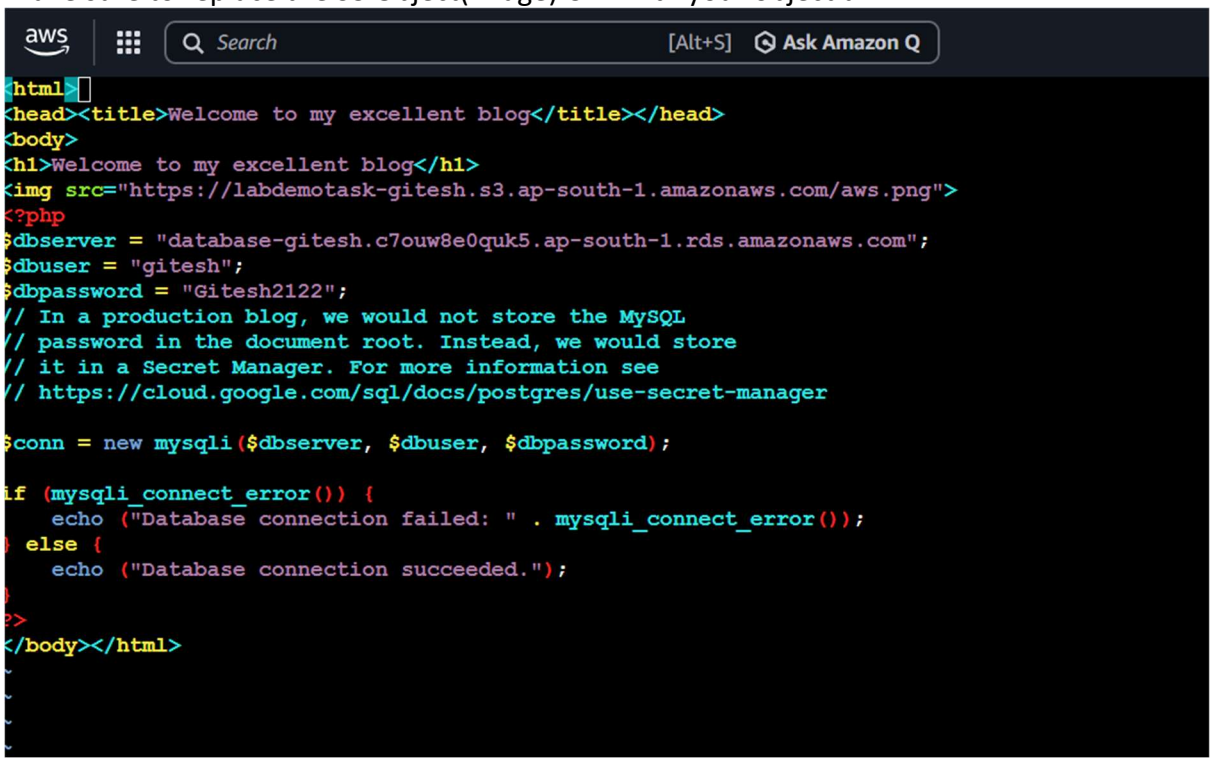



8. Connect to the instance created in task one. Give below command.

```
$ sudo su
$ cd /var/www/html
$ vi index.php
```



9. In the vi text editor, press “**insert**” key. Go to the end of the line which is ending with </h1> then press **Enter** key to add one blank line. Now paste below line in text editor.
<img src='S3 Object(image) URL'>
Make sure to replace the S3 Object(image) URL with your object url.

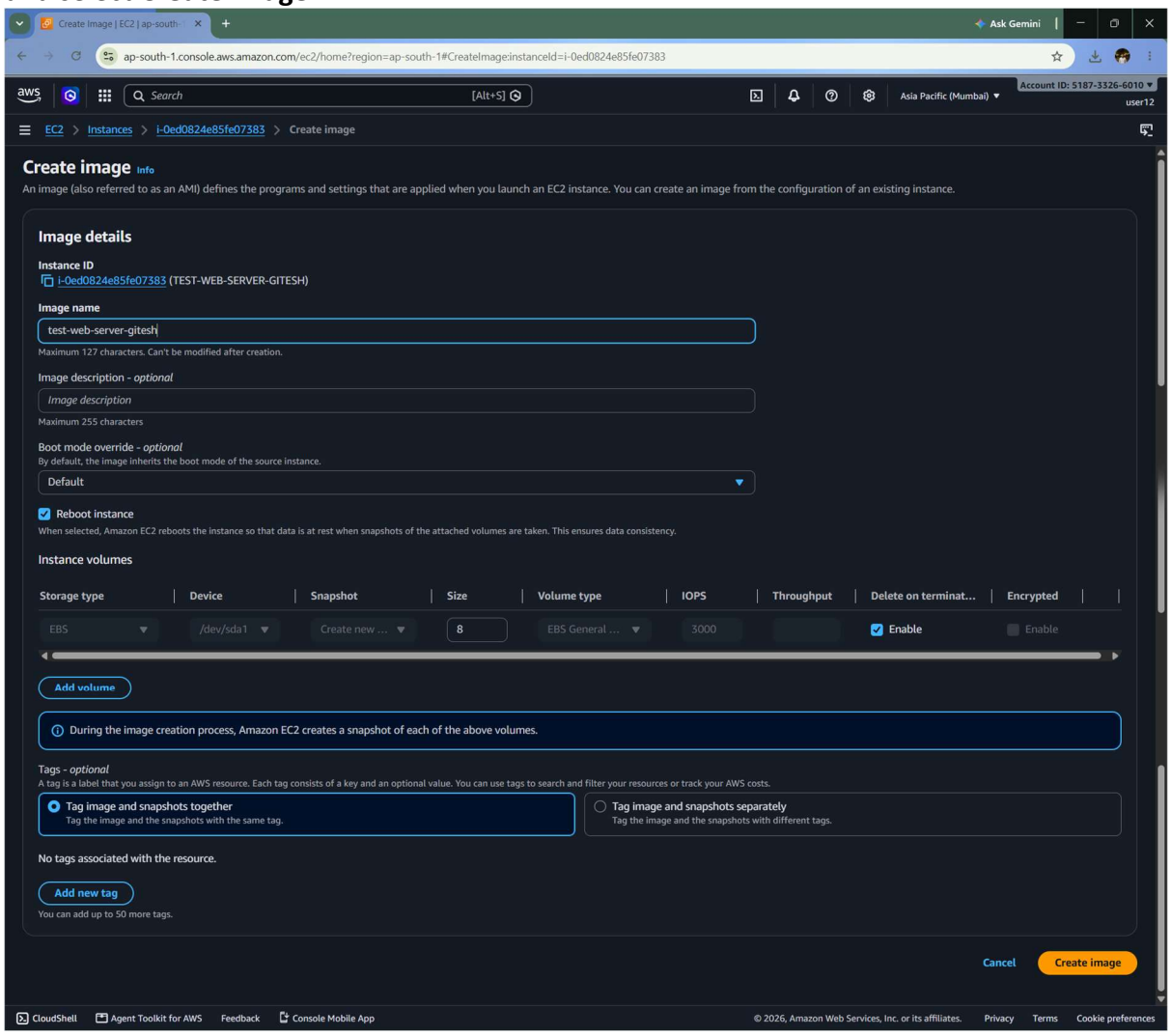



10. Now copy and paste the public IP of your instance in browser and add /index.php after the public IP of your instance. Below is the expected output. (Note that it is now showing the image pulled by ec2 instance from s3 buck and if you scroll down at the end on browser it displays database is connected successfully). It means that application server is connected to database server and s3 bucket also successfully.



# Task 7: Create AMI image of Instance.

1. Now your application is completely working so we will take AMI image of our server. For this go to EC2 service on console, select your instance, click on **Action**, and click on **Images and templates**. Click on **Create Image.**
2. In create Image window under **Image name** give any name, keep other option default and select **Create image.**

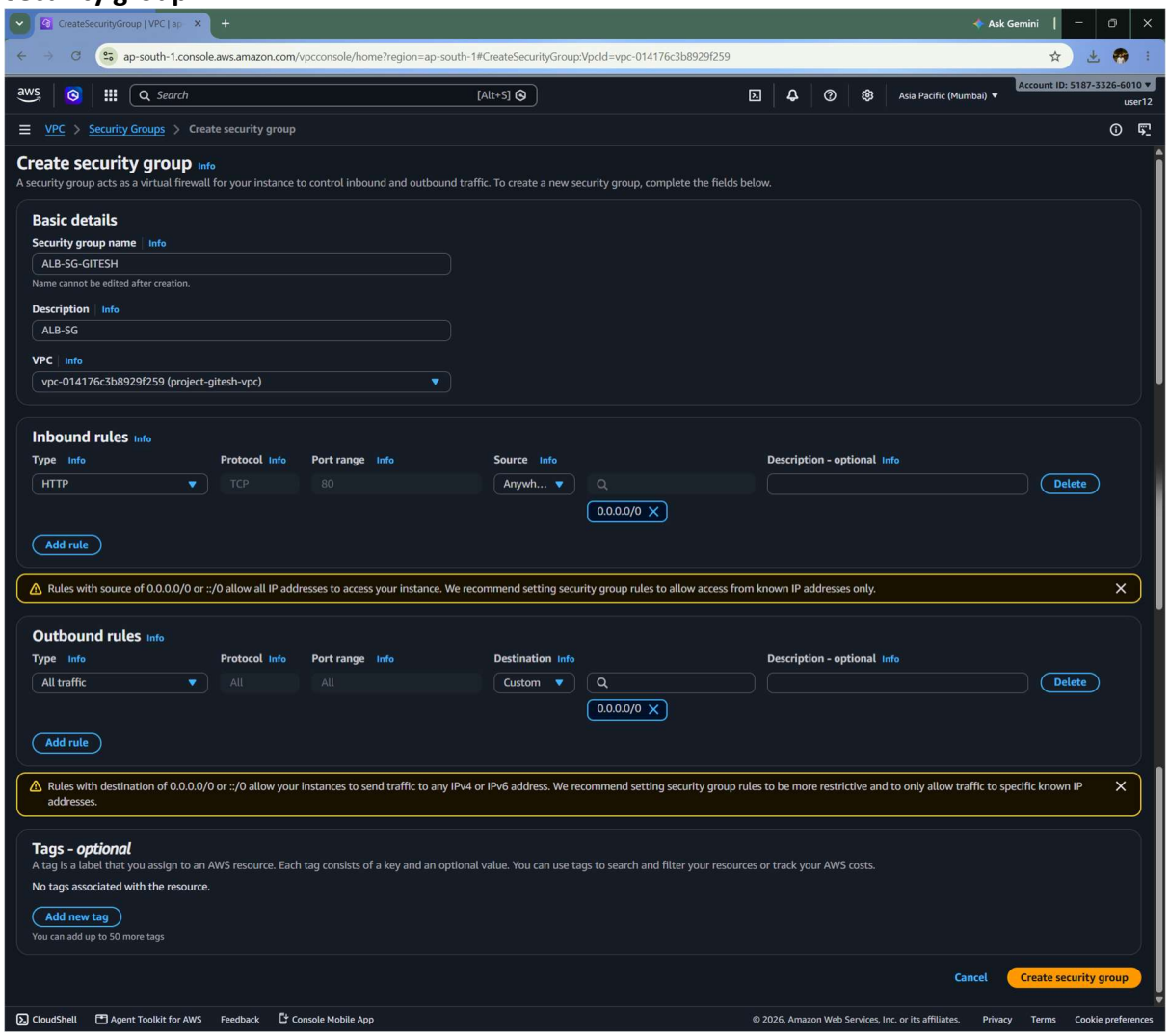


# Task 8: Create security group for ALB and modify security group of instance.

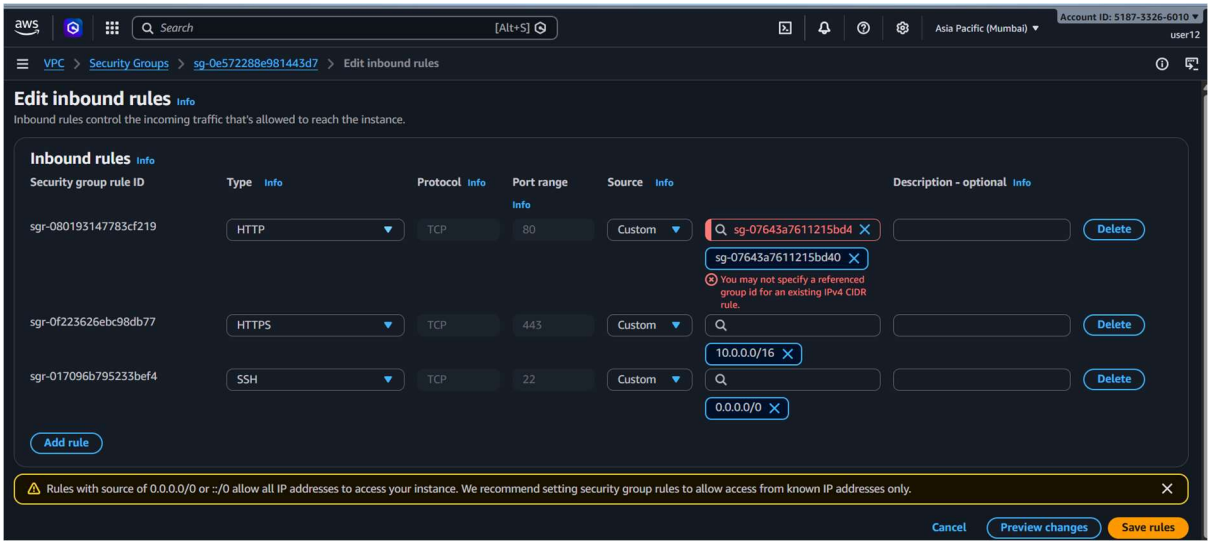
1. Go to the AWS Management console and search for VPC and in the left-hand side we will find the Security Group. Click on it.

2. Select **Create security group.**
3. Give name and description as ALB-SG and select demo-vpc created in step 1. In the inbound rule section click on **Add rule.**

4. Add inbound rule as shown below. Keep other option default and click on **Create security group.**

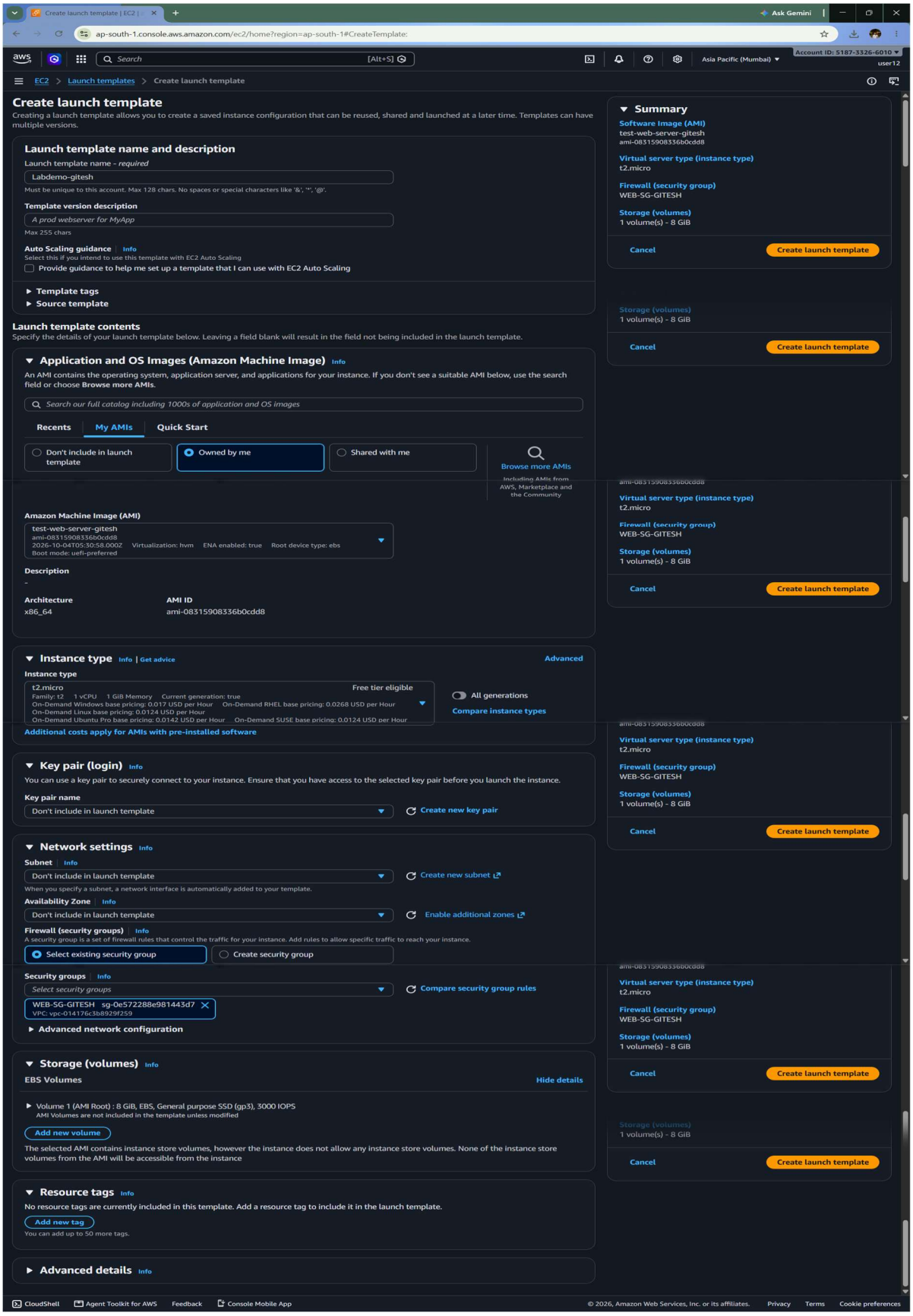


5. Now modify the security group created in task 2, step 3(security group attach to instance). Select WEB-SG security group, go to **Action, Edit inbound rule.**
6. Allow http traffic to ALB-SG security group. Allow SSH traffic to VPC CIDR. As shown below.

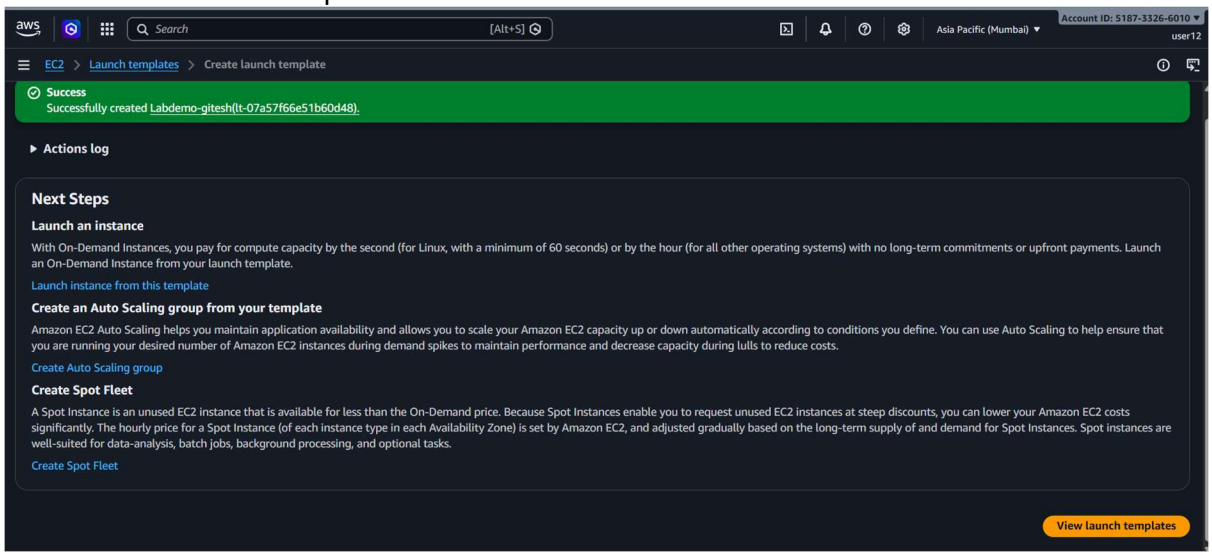


# Task 9: Implement high availability and scalability to the application.

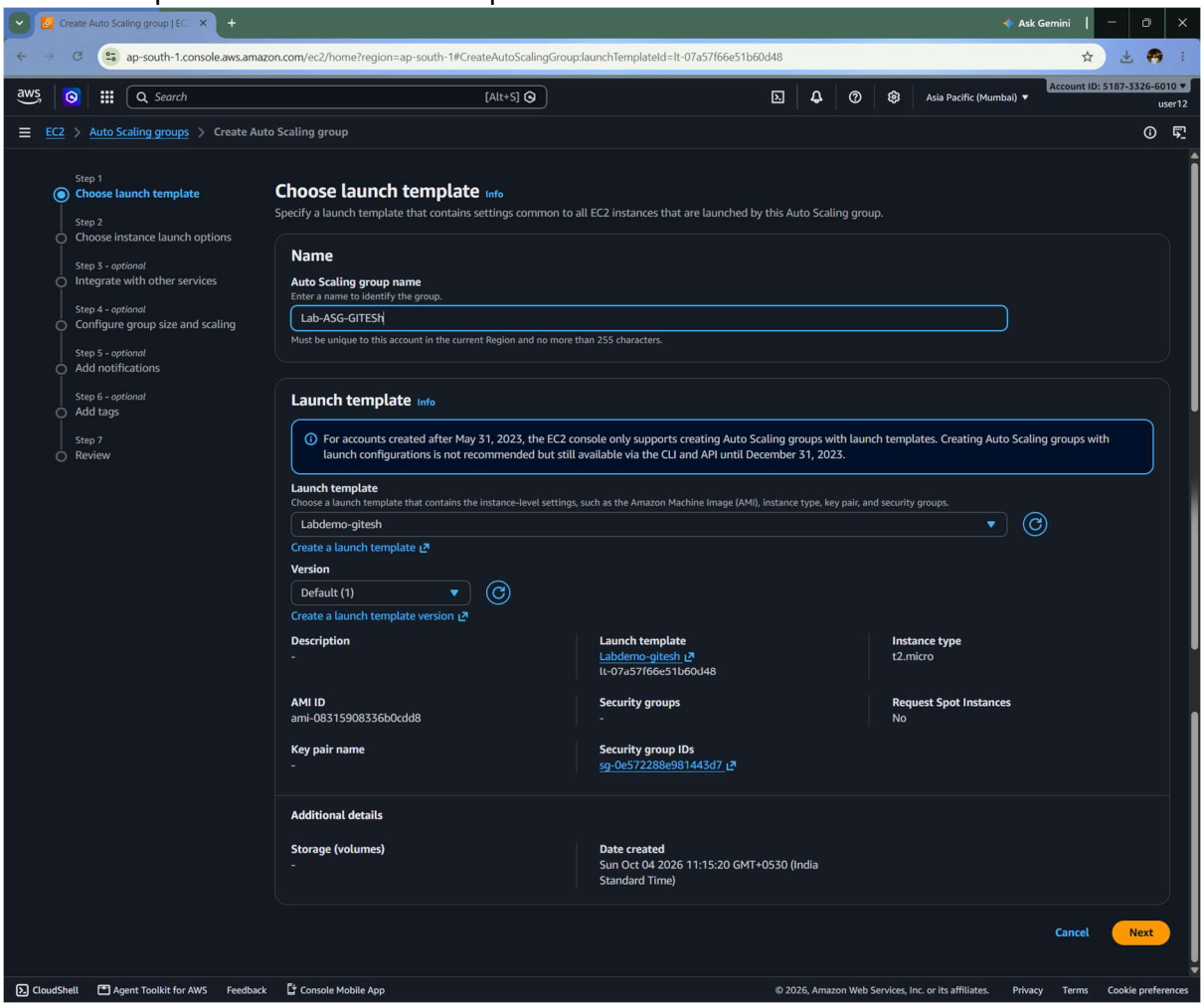
1. Create launch template-Go to **EC2 service,** select **launch template** from left-hand side and click on **create launch template**.
2. In create launch template window select below options:
    a. **Name**- Labdemo
    b. Under **Application and OS images** section select **My AMI**. Select the AMI created in task 7.
    c. **Instance Type**- t2.micro
    d. **Security Group**- WEB-SG
    e. **Key pair name-** Don't include in launch template.
    f. **Subnet-** Don't include in launch template.
    g. **Firewall- Select existing security group** then select WEB-SG
    h. Keep other option default and select **create launch template.**


Create launch template | EC2
Ask Gemini
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTemplate:
Search [Alt+S]
Asia Pacific (Mumbai)
Account ID: 5187-3326-6010
user12
EC2 > Launch templates > Create launch template
Create launch template
Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.
Launch template name and description
Launch template name - required
Labdemo-gitesh
Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.
Template version description
A prod webserver for MyApp
Max 255 chars
Auto Scaling guidance | Info
Select this if you intend to use this template with EC2 Auto Scaling
Provide guidance to help me set up a template that I can use with EC2 Auto Scaling
Template tags
Source template
Launch template contents
Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Don't include in launch template
Owned by me
Shared with me
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
test-web-server-gitesh
ami-08315908336b0cdd8
2026-10-04T05:30:58.000Z Virtualization: hvm ENA enabled: true Root device type: ebs
Boot mode: uefi-preferred
Description
-
Architecture
x86_64
AMI ID
ami-08315908336b0cdd8
Instance type Info | Get advice
Advanced
Instance type
t2.micro
Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name
Don't include in launch template
Create new key pair
Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone Info
Don't include in launch template
Enable additional zones
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Select existing security group
Create security group
Security groups Info
Select security groups
Compare security group rules
WEB-SG-GITESH sg-0e572288e981443d7
VPC: vpc-014176c3b8929f259
Advanced network configuration
Storage (volumes) Info
EBS Volumes
Hide details
Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified
Add new volume
The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance
Resource tags Info
No resource tags are currently included in this template. Add a resource tag to include it in the launch template.
Add new tag
You can add up to 50 more tags.
Advanced details Info
Summary
Software Image (AMI)
test-web-server-gitesh
ami-08315908336b0cdd8
Virtual server type (instance type)
t2.micro
Firewall (security group)
WEB-SG-GITESH
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
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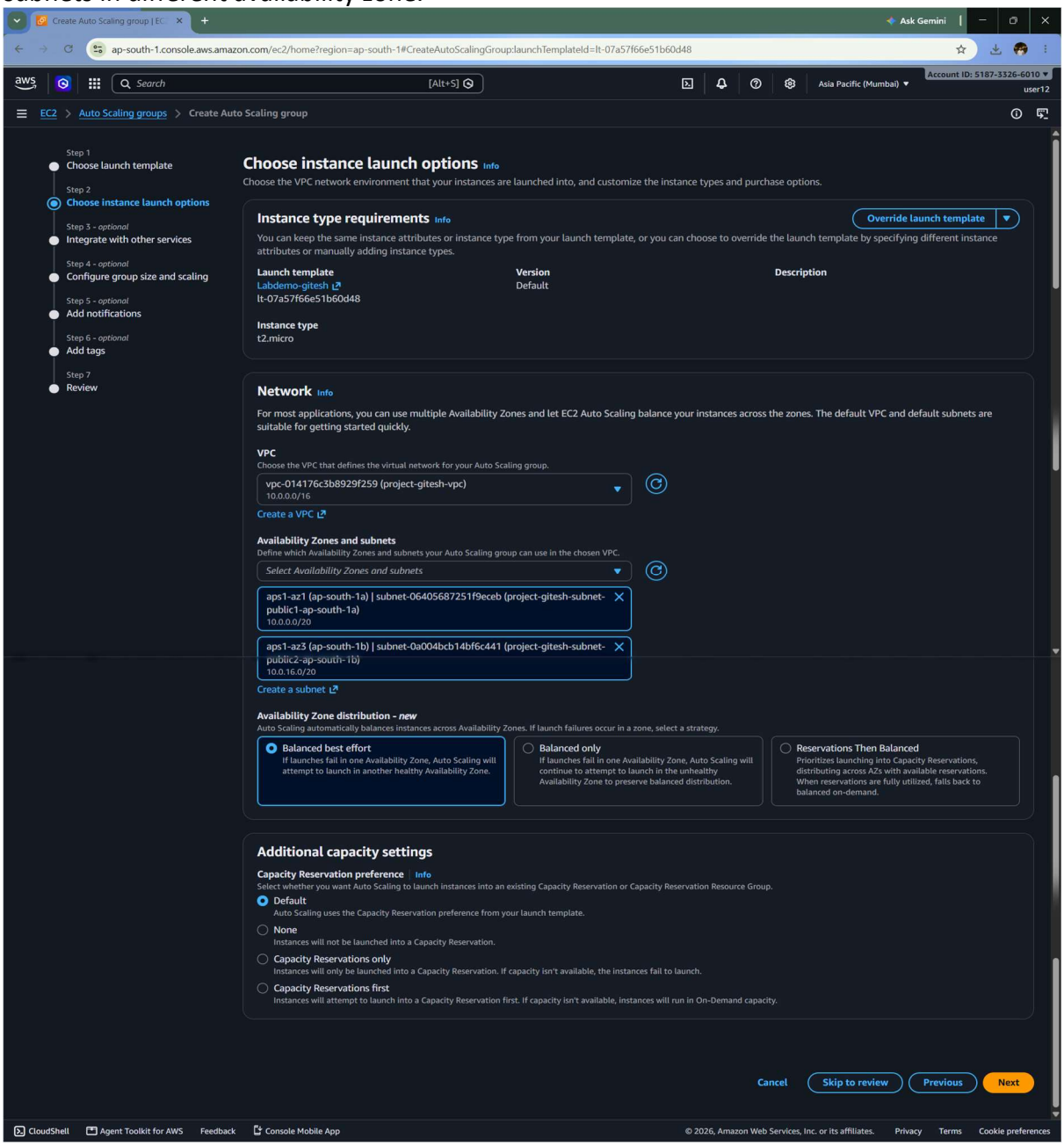
3. Click on view launch template.



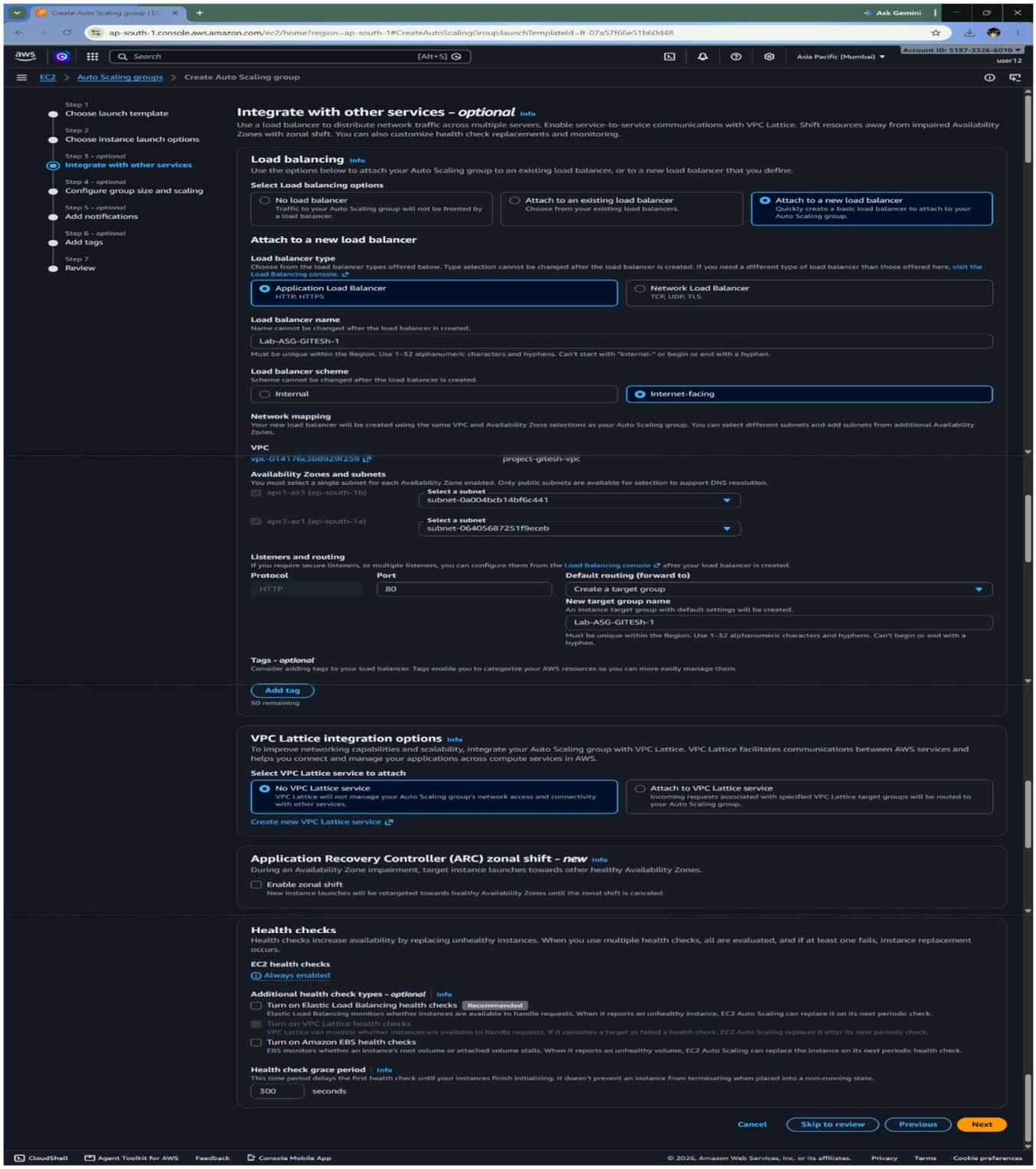

4. Select you template go to **Action** and click on **create autoscaling group**.
5. Under **Choose launch template or configuration** give name to autoscaling group, select the template created in above step. Click **Next.**



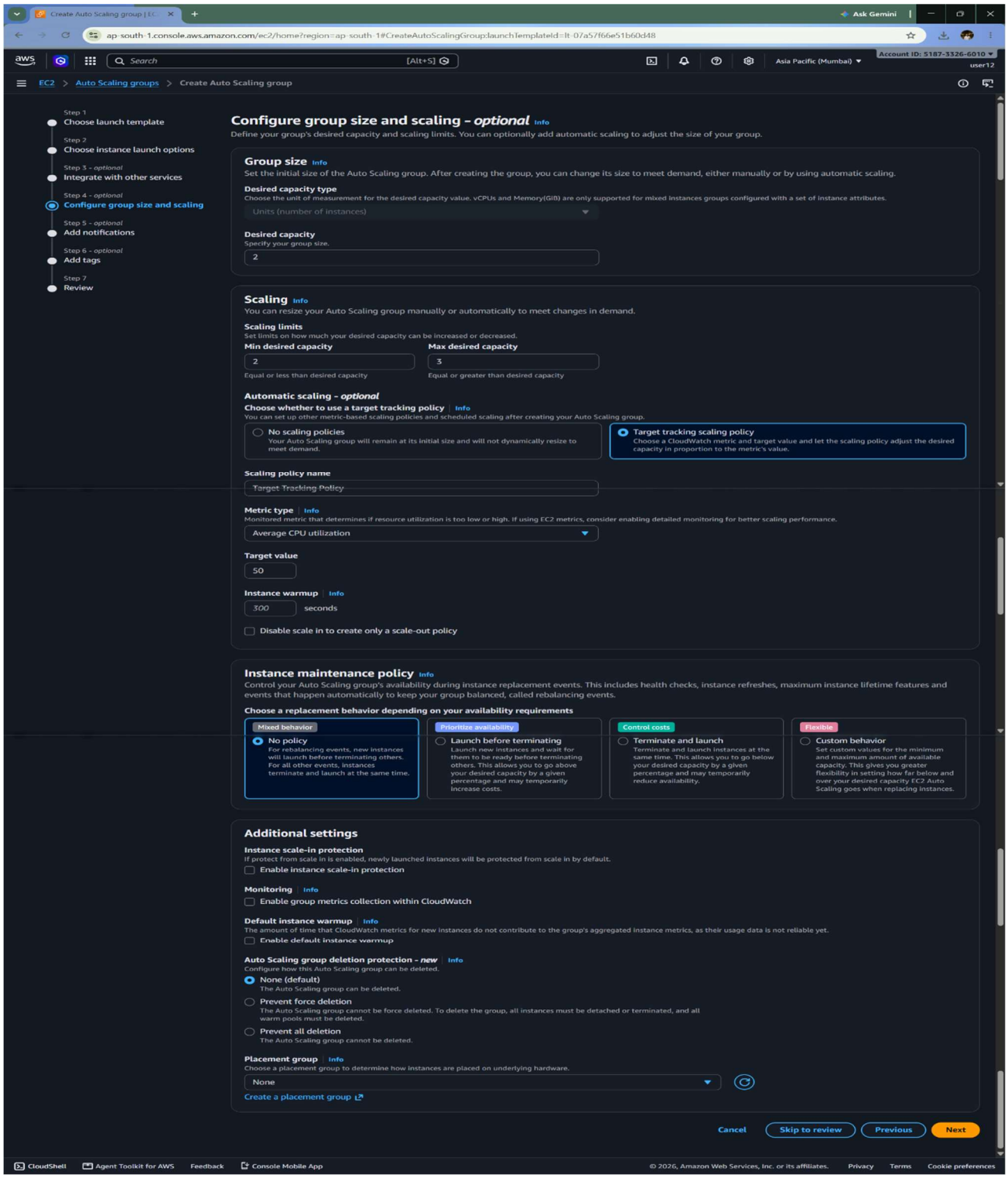
6. Under **Choose instance launch options** window select demo-vpc and select 2 public subnets in different availability zone.



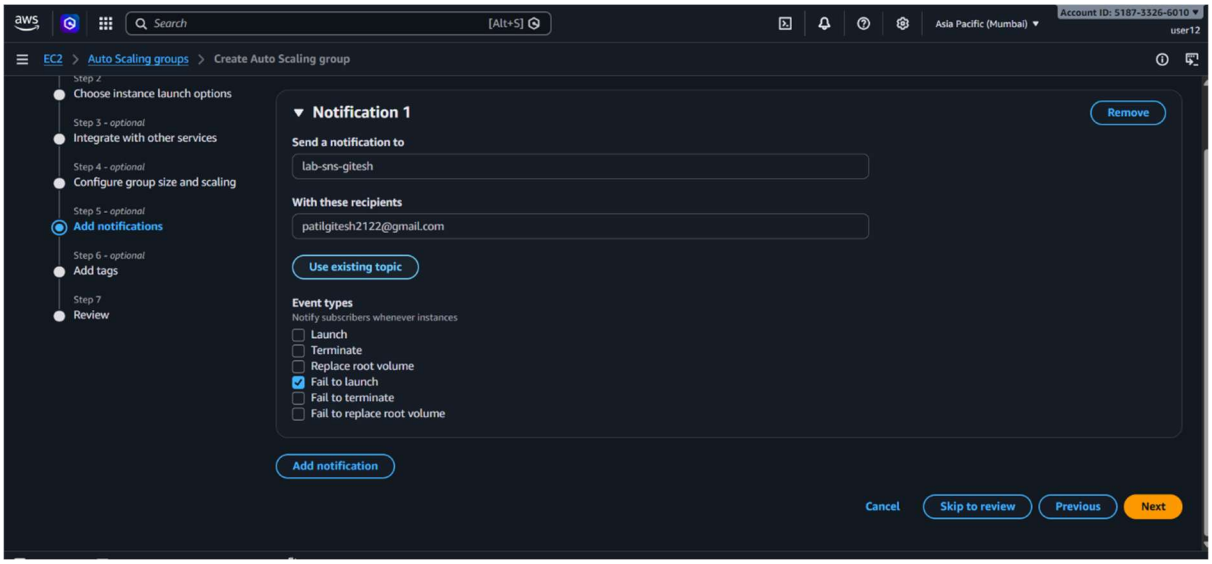

7. Under **Configure advanced options – optional** window selects the options as below:
    a. **Load Balancing**: Attach to a new load balancer.
    a) **Load Balancer Type**: Application Load Balancer.
    b) **Load Balancer Name**: Lab-ALB-1
    **c) Load Balancer Scheme: Internet-facing.**
    **d) In Listeners and routing under Default routing: Create a target group.**
    **e) In New target group name: Lab-TG-1.**
    **f) Click Next.**


Create Auto Scaling group | EC
Ask Gemini
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:launchTemplateId=lt-07a57f66e51b60d48
Search [Alt+S]
Asia Pacific (Mumbai)
Account ID: 5187-3326-6010
user12
EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1 Choose launch template
Step 2 Choose instance launch options
Step 3 - optional Integrate with other services
Step 4 - optional Configure group size and scaling
Step 5 - optional Add notifications
Step 6 - optional Add tags
Step 7 Review
Integrate with other services – optional Info
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
Select Load balancing options
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.
Attach to a new load balancer
Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the Load Balancing console.
Application Load Balancer
HTTP, HTTPS
Network Load Balancer
TCP, UDP, TLS
Load balancer name
Name cannot be changed after the load balancer is created.
Lab-ASG-GITESh-1
Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't start with "internal-" or begin or end with a hyphen.
Load balancer scheme
Scheme cannot be changed after the load balancer is created.
Internal
Internet-facing
Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.
VPC
vpc-014176c5b8929f259
project-gitesh-vpc
Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.
aps1-az3 (ap-south-1b)
Select a subnet
subnet-0a004bcb14bf6c441
aps1-az1 (ap-south-1a)
Select a subnet
subnet-06405687251f9eceb
Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the Load Balancing console after your load balancer is created.
Protocol
HTTP
Port
80
Default routing (forward to)
Create a target group
New target group name
An instance target group with default settings will be created.
Lab-ASG-GITESh-1
Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't begin or end with a hyphen.
Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.
Add tag
50 remaining
VPC Lattice integration options Info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.
Select VPC Lattice service to attach
No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.
Create new VPC Lattice service
Application Recovery Controller (ARC) zonal shift - new Info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.
Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
EC2 health checks
Always enabled
Additional health check types - optional Info
Turn on Elastic Load Balancing health checks Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.
Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.
Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.
Health check grace period Info
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.
300
seconds
Cancel
Skip to review
Previous
Next
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

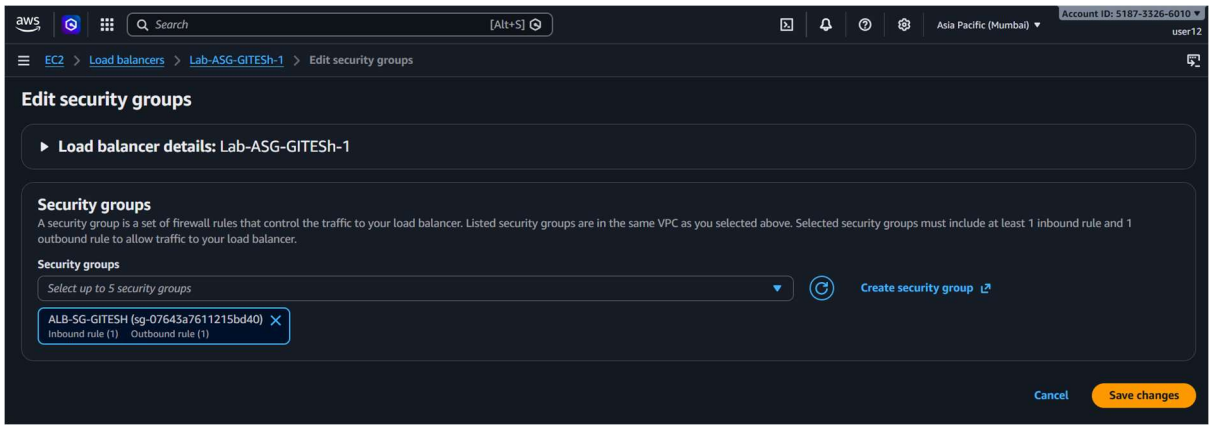
8. Under **Configure group size and scaling – optional** window selects the options as below:
    a. **In group size**: Desired capacity set in 2.
    b. **Scaling**: Min desired capacity set to 2.
        Max desired capacity set to 3.
    c. In **Automatic scaling** – optional: Select Target tracking scaling policy. Keepother option default.
    d. Click **Next**



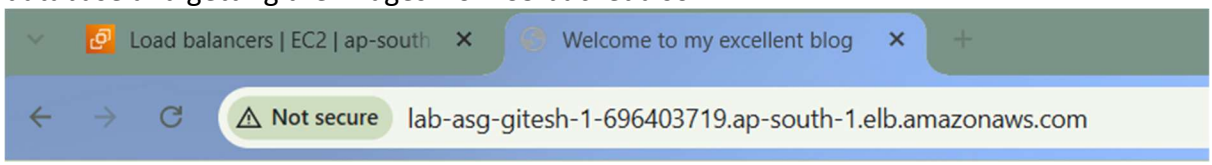
9. In **Add notifications – optional** window select **Add notifications.** Then select below options:
    a. Create a topic.
    b. Send a notification: lab-sns
    c. With these recipients: Give your personal mail-id.
    d. In event type: Enable Fail to Launch and disable all remaining 3 options.



10. In **Add tags -Optional window** click next.
11. In **Review** window click **create autoscaling group.**
12. Now this auto scaling group will create load balancer, target group, 2 instances. But load balancer will come up with security group of WEB-SG. We should change this to ALB-SG security group. For this go to EC2 service from left-hand side select **Load balancer**
13. Now select the load balancer created by autoscaling group, go to **Action** and click on **Edit security group.**
14. Under Security group select ALB-SG and disable WEB-SG. Then click on **Save changes**.




15. Copy load balancer dns name and paste it in new tab of browser. Add /index.php after load balancer dns name.
16. Below is the expected output which shows webserver successfully connected to RDS database and getting the images from S3 bucket also.



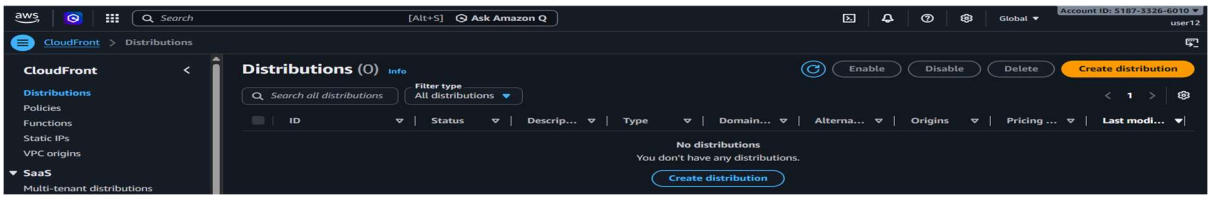

# Welcome to my excellent blog



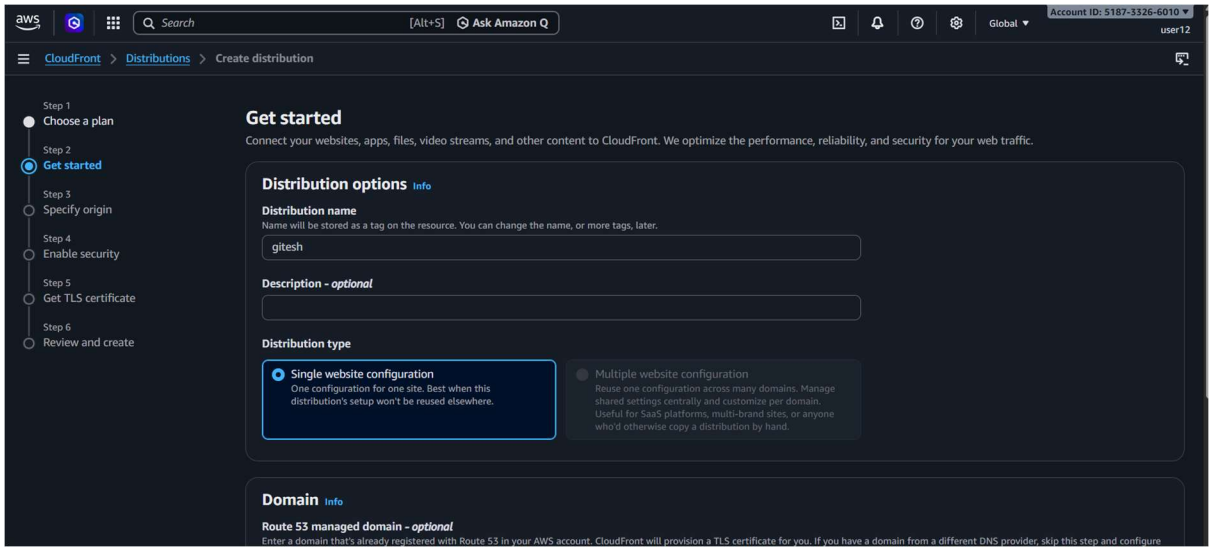

Database connection succeeded.

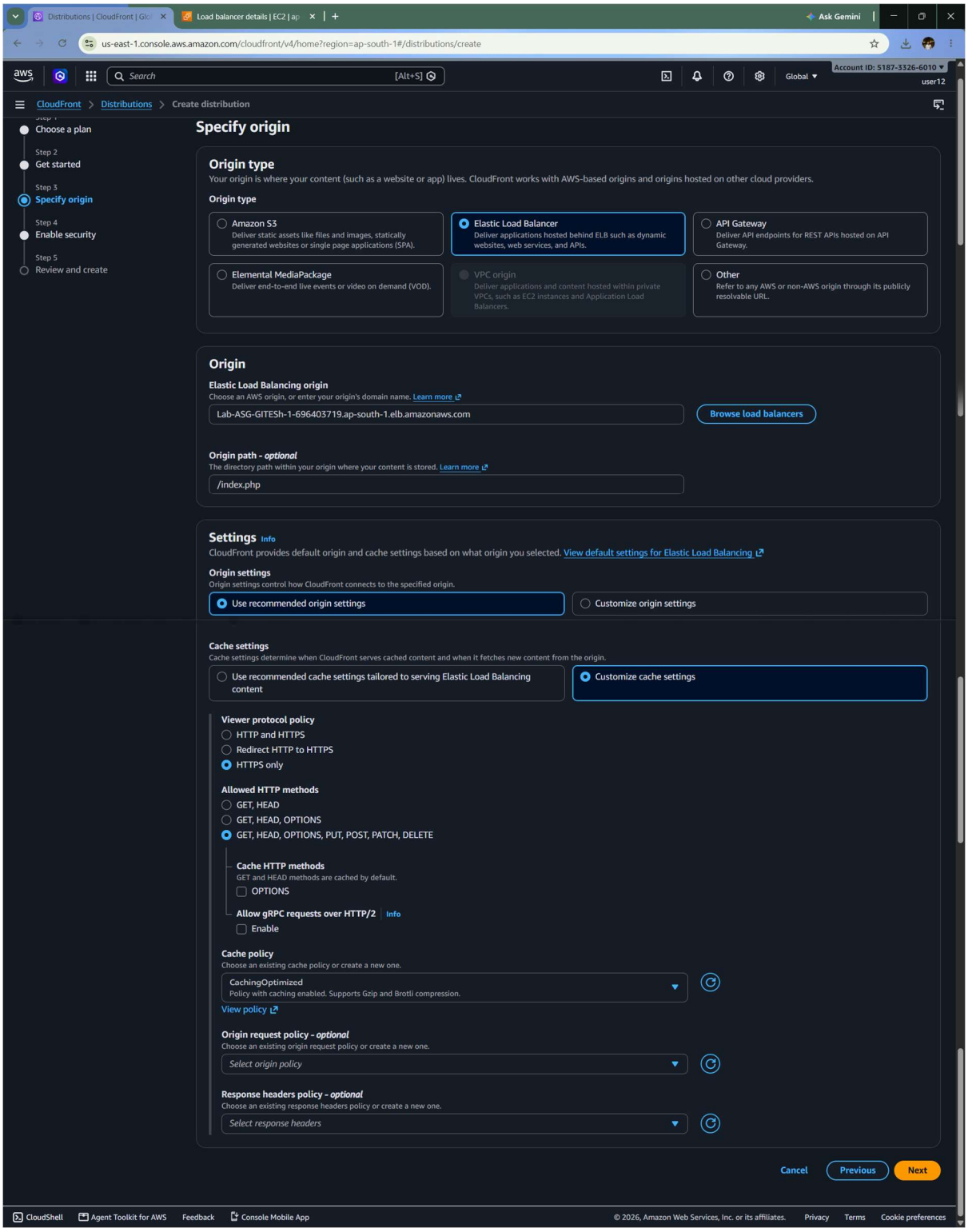
# Task 10: Implement low latency and security from DDoS attack to the application.

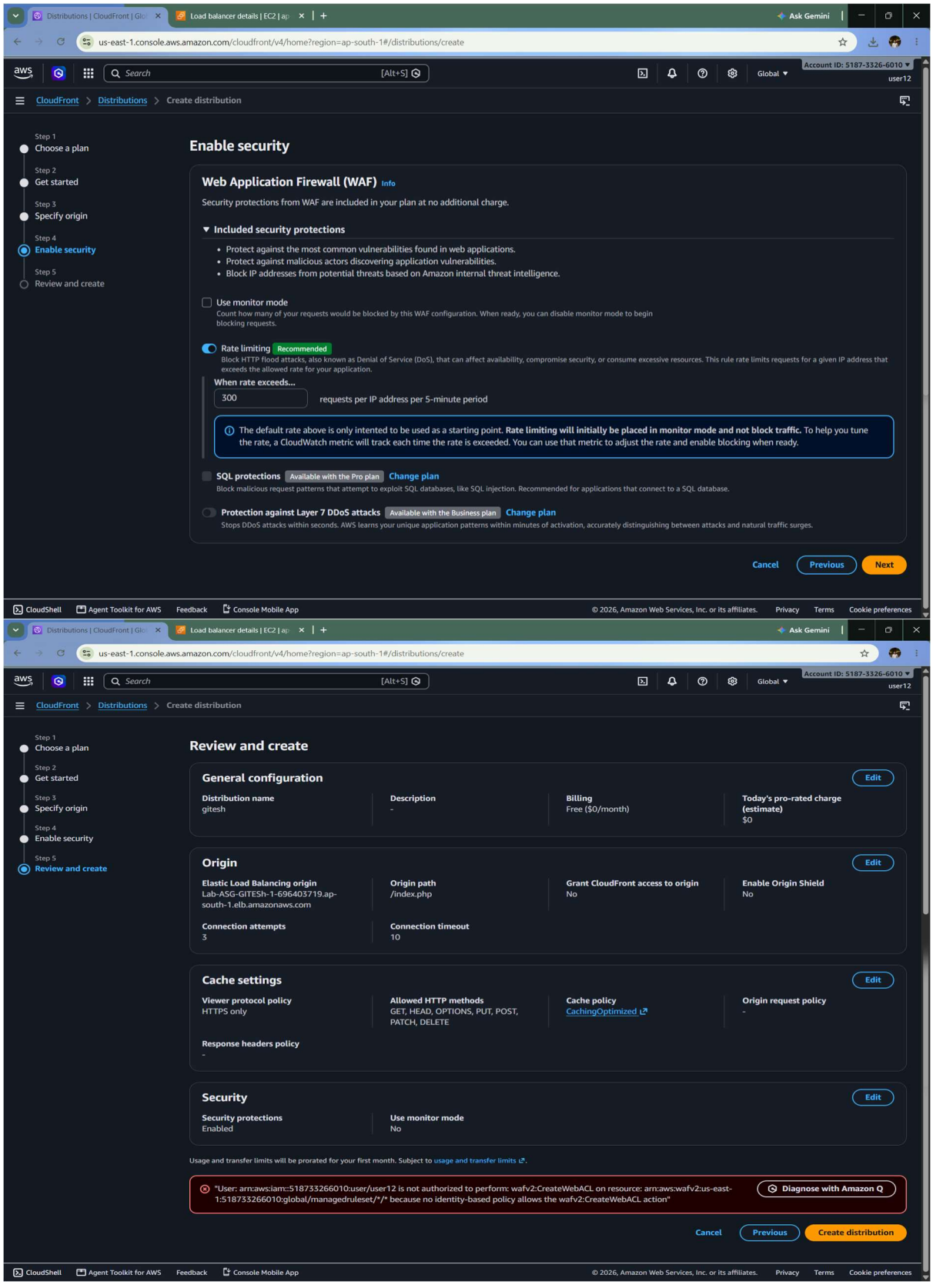
1. Go to CloudFront service on console and click on create distributions.




2. In **origin** section, select your created load balancer as origin, in **origin path** add /index.php.
3. In **Caching policy** section select caching optimized policy
4. In Web Application Firewall section select Enable security protections, enable SQL protections, enable Rate limiting.
5. In Default root object section write index.php, click on Create distribution.




Distributions | CloudFront | Glo
Load balancer details | EC2 | ap
Ask Gemini
us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=ap-south-1#/distributions/create
Search
[Alt+S]
Global
Account ID: 5187-3326-6010
user12
CloudFront > Distributions > Create distribution
Choose a plan
Step 2
Get started
Step 3
Specify origin
Step 4
Enable security
Step 5
Review and create
Specify origin
Origin type
Your origin is where your content (such as a website or app) lives. CloudFront works with AWS-based origins and origins hosted on other cloud providers.
Origin type
Amazon S3
Deliver static assets like files and images, statically generated websites or single page applications (SPA).
Elastic Load Balancer
Deliver applications hosted behind ELB such as dynamic websites, web services, and APIs.
API Gateway
Deliver API endpoints for REST APIs hosted on API Gateway.
Elemental MediaPackage
Deliver end-to-end live events or video on demand (VOD).
VPC origin
Deliver applications and content hosted within private VPCs, such as EC2 instances and Application Load Balancers.
Other
Refer to any AWS or non-AWS origin through its publicly resolvable URL.
Origin
Elastic Load Balancing origin
Choose an AWS origin, or enter your origin's domain name. Learn more
Lab-ASG-GITESh-1-696403719.ap-south-1.elb.amazonaws.com
Browse load balancers
Origin path - optional
The directory path within your origin where your content is stored. Learn more
/index.php
Settings Info
CloudFront provides default origin and cache settings based on what origin you selected. View default settings for Elastic Load Balancing
Origin settings
Origin settings control how CloudFront connects to the specified origin.
Use recommended origin settings
Customize origin settings
Cache settings
Cache settings determine when CloudFront serves cached content and when it fetches new content from the origin.
Use recommended cache settings tailored to serving Elastic Load Balancing content
Customize cache settings
Viewer protocol policy
HTTP and HTTPS
Redirect HTTP to HTTPS
HTTPS only
Allowed HTTP methods
GET, HEAD
GET, HEAD, OPTIONS
GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE
Cache HTTP methods
GET and HEAD methods are cached by default.
OPTIONS
Allow gRPC requests over HTTP/2 Info
Enable
Cache policy
Choose an existing cache policy or create a new one.
CachingOptimized
Policy with caching enabled. Supports Gzip and Brotli compression.
View policy
Origin request policy - optional
Choose an existing origin request policy or create a new one.
Select origin policy
Response headers policy - optional
Choose an existing response headers policy or create a new one.
Select response headers
Cancel
Previous
Next
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Distributions | CloudFront | Glo
Load balancer details | EC2 | ap
Ask Gemini
us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=ap-south-1#/distributions/create
Search [Alt+S]
Global
Account ID: 5187-3326-6010
user12
CloudFront > Distributions > Create distribution
Step 1 Choose a plan
Step 2 Get started
Step 3 Specify origin
Step 4 Enable security
Step 5 Review and create
Enable security
Web Application Firewall (WAF) Info
Security protections from WAF are included in your plan at no additional charge.
Included security protections
Protect against the most common vulnerabilities found in web applications.
Protect against malicious actors discovering application vulnerabilities.
Block IP addresses from potential threats based on Amazon internal threat intelligence.
Use monitor mode
Count how many of your requests would be blocked by this WAF configuration. When ready, you can disable monitor mode to begin blocking requests.
Rate limiting Recommended
Block HTTP flood attacks, also known as Denial of Service (DoS), that can affect availability, compromise security, or consume excessive resources. This rule rate limits requests for a given IP address that exceeds the allowed rate for your application.
When rate exceeds...
300
requests per IP address per 5-minute period
The default rate above is only intented to be used as a starting point. Rate limiting will initially be placed in monitor mode and not block traffic. To help you tune the rate, a CloudWatch metric will track each time the rate is exceeded. You can use that metric to adjust the rate and enable blocking when ready.
SQL protections Available with the Pro plan Change plan
Block malicious request patterns that attempt to exploit SQL databases, like SQL injection. Recommended for applications that connect to a SQL database.
Protection against Layer 7 DDoS attacks Available with the Business plan Change plan
Stops DDoS attacks within seconds. AWS learns your unique application patterns within minutes of activation, accurately distinguishing between attacks and natural traffic surges.
Cancel
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Distributions | CloudFront | Glo
Load balancer details | EC2 | ap
Ask Gemini
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Search [Alt+S]
Global
Account ID: 5187-3326-6010
user12
CloudFront > Distributions > Create distribution
Step 1 Choose a plan
Step 2 Get started
Step 3 Specify origin
Step 4 Enable security
Step 5 Review and create
Review and create
General configuration
Edit
Distribution name
gitesh
Description
-
Billing
Free ($0/month)
Today's pro-rated charge (estimate)
$0
Origin
Edit
Elastic Load Balancing origin
Lab-ASG-GITESh-1-696403719.ap-south-1.elb.amazonaws.com
Origin path
/index.php
Grant CloudFront access to origin
No
Enable Origin Shield
No
Connection attempts
3
Connection timeout
10
Cache settings
Edit
Viewer protocol policy
HTTPS only
Allowed HTTP methods
GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE
Cache policy
CachingOptimized
Origin request policy
-
Response headers policy
-
Security
Edit
Security protections
Enabled
Use monitor mode
No
Usage and transfer limits will be prorated for your first month. Subject to usage and transfer limits.
"User: arn:aws:iam::518733266010:user/user12 is not authorized to perform: wafv2:CreateWebACL on resource: arn:aws:wafv2:us-east-1:518733266010:global/managedruleset/*/* because no identity-based policy allows the wafv2:CreateWebACL action"
Diagnose with Amazon Q
Cancel
Previous
Create distribution
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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